MATH 315 (Winter 2026)

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Chapter 1

Introduction

1.1 A simple first-order ODE

An ordinary differential equation (ODE) is an equation involving a function and its derivatives.

A simple first-order ODE has the form:

$$y' = \frac{dy}{dx} = ky \quad k \in \mathbb{R},\ k \in \mathbb{C}$$

To solve for the function y, we can divide both sides by y then integrate both sides:

$$\begin{aligned}
\int \frac{dy}{dx} \cdot \frac{1}{y} dx &= \int k\ dx \\
\ln|y| &= kx + C \\
y &= e^{kx+C} \\
y &= e^{kx} e^{C} \\
\therefore \quad & \boxed{y = Ce^{kx}}
\end{aligned}$$

If we apply an **initial condition**, such as $y(0) = y_0$, we can solve for the constant C:

$$\begin{aligned}
y(0) &= Ce^{k \cdot 0} = C = y_0 \\
\therefore y &= y_0 e^{kx}
\end{aligned}$$

Example 1.1.1

Find the general solution of $\frac{dy}{dx} = 2y + 1$.

Solution: We can rearrange the equation as follows:

$$\begin{aligned}
\frac{dy}{dx} \cdot \frac{1}{2y+1} &= 1 \\
\int \frac{1}{2y+1}\ dy &= \int 1\ dx \\
\frac{1}{2} \ln|2y+1| &= x + C \\
\ln|2y+1| &= 2x + C' \\
2y + 1 &= e^{2x+C'} \\
2y + 1 &= Ce^{2x} \\
y &= \frac{Ce^{2x} - 1}{2}
\end{aligned}$$

We can verify this solution by differentiating it:

$$\frac{dy}{dx} = \frac{1}{2} \cdot C \cdot 2e^{2x} = Ce^{2x}$$
$$2y + 1 = 2 \cdot \frac{Ce^{2x} - 1}{2} + 1 = Ce^{2x}$$
$$\therefore \frac{dy}{dx} = 2y + 1$$

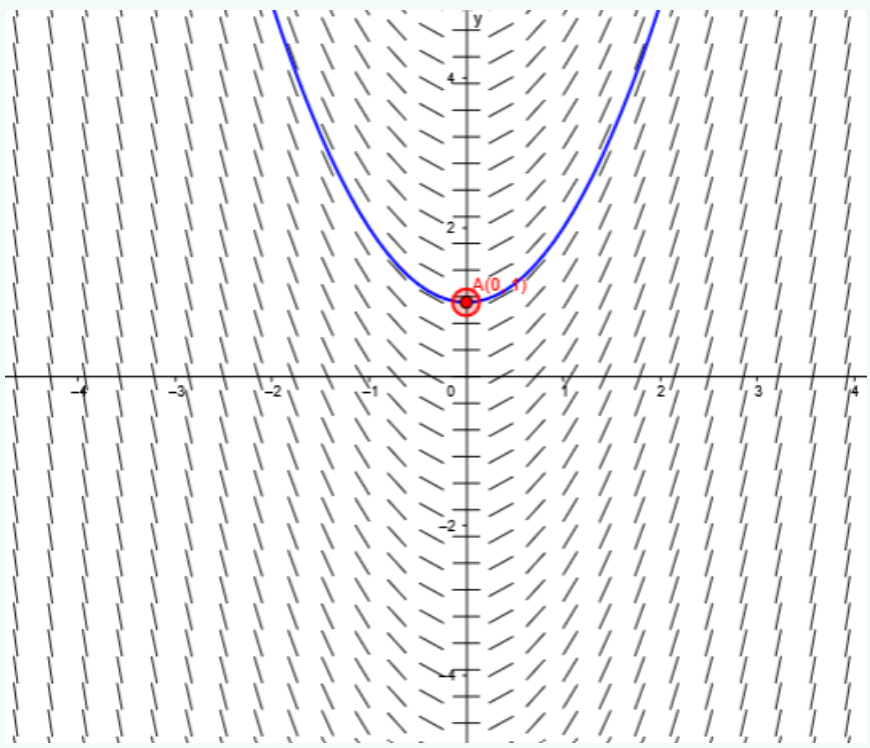
1.2 Graphical interpretation of a first-order ODE

Given some equation of the form $y' = F(x, y)$, we could read this as "the slope of the function y at the point (x, y) is given by $F(x, y)$".

If $F(x, y) > 0$, then the function is increasing at that point; if $F(x, y) < 0$, then the function is decreasing at that point. At every point of the xy-plane, we can compute the slope of the solution curve that passes through that point. We call this the **slope field** of the differential equation.

Example 1.2.1

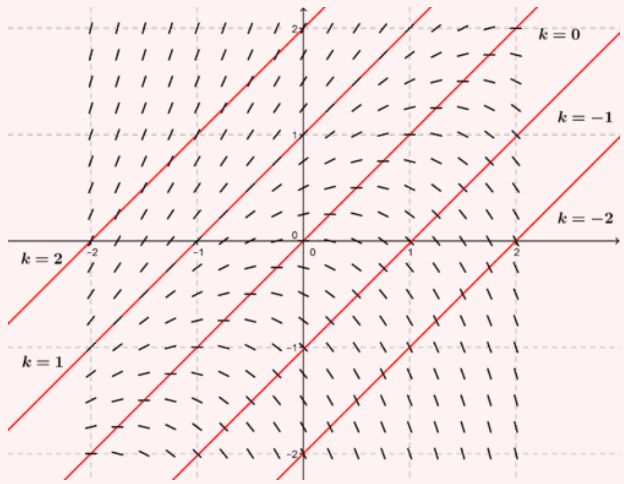
Given the differential equation $\frac{dy}{dx} = 2x$, we can draw a slope field by evaluating the slope at every (x, y) point.
This produces a slope field that looks like this:



The general solution of this differential equation is $y = x^2 + C$.
We can draw a curve that is tangential to the slopes at every point to find a solution curve (or **integral curve**).
If we pick an initial condition, such as $y(0) = 1$, we can approximate the solution curve by following the slopes in the slope field.

Definition 1.2.1: Isoclines

Isoclines are curves along which the slope of the solution curve is constant.
For the differential equation $\frac{dy}{dx} = F(x, y)$, the isocline for slope m is given by the equation $F(x, y) = m$.



1.3 Numerical approximation

1.3.1 Euler's method

Most differential equations cannot be solved analytically.

We define a function $f(x) = \dot{x}$, where $\dot{x}$ is the derivative of x with respect to t.

Recall the limit definition of the derivative:

$$\dot{x} = \lim_{h \to 0} \frac{x(t+h) - x(t)}{h}$$

Then for some sufficiently small h, we can approximate:

$$\dot{x} \approx \frac{x(t+h) - x(t)}{h}$$

Since we have that $\dot{x} = f(x)$, we can rearrange this to get:

$$\boxed{x(t+h) \approx x(t) + hf(x(t))}$$

This is known as **Euler's method**.

We can also write it in discrete form as:

$$\boxed{x_{m+1} = x_m + hf(x_m)}$$

where $x_m = x(t_m)$ and $t_m = t_0 + mh$.

The Euler backward method looks like:

$$\boxed{x_{m+1} = x_m + hf(x_{m+1})}$$

However, in practice, it is often difficult to solve for x_{m+1} in this equation. Rather, the improved Euler method uses the average of the slopes at the beginning and end of the interval. Then using our approximation of x_{m+1} from the Euler forward method, we could average the "beginning" of the step and the "end" of the step as follows:

$$\tilde{x}_{m+1} = x_m + hf(x_m)$$
$$x_{m+1} = x_m + \frac{1}{2}h\left(f(x_m) + f(\tilde{x}_{m+1})\right)$$

$$\boxed{x_{m+1} = x_m + \frac{h}{2}\left(f(x_m) + f\left(x_m + hf(x_m)\right)\right)}$$

1.3.2 Local and global error

The **global error** could be written as:

$$E = |x(t_N) - x_N|$$

where $x(t_N)$ is the exact solution at time t_N and x_N is the numerical approximation at time t_N.
Let the final time be $T = N \cdot h$, where h is the step size and N is the number of steps taken.
The **local error** at step m is defined as:

$$E_m = |x(t_m) - x_m|$$

To analyze the error, we can use Taylor's theorem to expand $x(t_{m+1})$ around t_m:

$$x(t_{m+1}) = x(t_m) + h\dot{x}(t_m) + \frac{h^2}{2}\ddot{x}(t_m)$$

Then for **Euler's forward method**, we have:

$$x(t_{m+1}) = x(t_m) + hf(x(t_m)) + e$$

Then we can bound e using the second derivative term:

In other words, by Taylor's theorem, the error e at each step is given by the h^2 term involving the second derivative, plus even smaller terms involving higher derivatives and higher powers of h. That is,

$$|e| \leqslant (\text{a constant}) \times h^2 \times \max_{t\in[0,T]} |\ddot{x}(t)| + (\text{smaller terms involving higher derivatives and higher powers of } h)$$

where the constant is $1/2$ in this case for the h^2 term.

So e (the local error) is dominated by the h^2 term, but in reality, there are also even smaller contributions from higher derivatives (like $\dddot{x}$) and higher powers of h.

To bound the global error, we add up the local errors over all N steps. This means the total (global) error E.

$$\begin{aligned} E &\leqslant \sum_{i=1}^{N} |e_i| \\ &\leqslant \frac{T}{h} \cdot h^2 \cdot \max_{t\in[0,T]} |\ddot{x}(t)| \\ &= T \cdot h \cdot \max_{t\in[0,T]} |\ddot{x}(t)| \end{aligned}$$

So the global error E is proportional to h for Euler's forward method.

Definition 1.3.1

If a numerical method has a one-step error of size h^{p+1}, then we say the method is of order p.
The global error is then of size h^p.

Chapter 2

Techniques for solving first-order ODEs

2.1 Separation of variables

The standard form of a 1st order ODE is given by:

$$\overline{R}(t)\dot{x} + \overline{P}(t)x = \overline{q}(t) \quad \overline{R}(t) \neq 0$$

If we let:

$$p = \frac{\overline{P}(t)}{\overline{R}(t)} \quad , \quad q = \frac{\overline{q}(t)}{\overline{R}(t)}$$

Then we obtain the equation in the form:

$$\boxed{\dot{x} + p(t)x = q(t)}$$

Definition 2.1.1: Homogeneous equation

An ODE is said to be **homogeneous** if $q(t) = 0$ for all t in the interval of interest.

Intuitively, the system evolves with no additional forcing or input beyond its initial conditions.
A homogeneous equation is also **separable** if it can be expressed in the form:

$$\frac{\dot{x}}{x} = f(t)$$

Definition 2.1.2: Separable equation

An ODE is said to be **separable** if it can be expressed in the form:

$$\frac{dx}{dt} = g(t)h(x)$$

where $g(t)$ is a function of t only and $h(x)$ is a function of x only.

We can easily solve separable equations by separating the variables and integrating both sides.

Example 2.1.1

Solve the following ODE:

$$\dot{x} + p(t)x = 0$$

Solution: We can rewrite the equation as:

$$\frac{\dot{x}}{x} = -p(t)$$

Then we separate the variables:

$$\frac{1}{x}\frac{dx}{dt} = -p(t)$$
$$\frac{1}{x}dx = -p(t)dt$$

Now we integrate both sides:

$$\int \frac{1}{x}dx = \int -p(t)dt$$
$$\ln|x| = -\int p(t)dt + C$$
$$|x| = e^{-\int p(t)dt + C}$$
$$x(t) = \pm e^{C} e^{-\int p(t)dt}$$
$$x(t) = Ke^{-\int p(t)dt} \quad , \quad K = \pm e^{C}$$

Thus, the general solution to the ODE is:

$$\boxed{x(t) = Ke^{-\int p(t)dt}}$$

2.1.1 Steps to solve separable equations

1. Rewrite the ODE in the form $\frac{dx}{dt} = g(t)h(x)$.
2. Separate the variables to obtain $\frac{1}{h(x)}dx = g(t)dt$.
3. Integrate both sides: $\int \frac{1}{h(x)}dx = \int g(t)dt$.
4. Solve for $x(t)$.
5. (Optional) Apply any initial conditions to solve for constants of integration.

Definition 2.1.3: Homogeneous solution

In general, the solution to a homogeneous ODE is called the **homogeneous solution**.

$$\boxed{X_h(t) = e^{-\int p(t)dt}}$$

Then:

$$x(t) = KX_h(t)$$

for some constant K.

Example 2.1.2 (Newton's law of cooling)

The change in temperature u over time t is proportional to the difference in temperature between the object and its surroundings.
Let u_0 be the initial temperature of the object and T_{ext} be the external temperature. Let k be the proportionality constant. Then:

$$\dot{u} = -k(u - T_{\text{ext}})$$

Solution: We can rewrite the equation as:

$$\dot{u} + ku = kT_{\text{ext}}$$

This is a first-order linear ODE. We can separate the variables:

$$\begin{aligned} \frac{du}{dt} &= -k(u - T_{\text{ext}}) \\ \frac{1}{u - T_{\text{ext}}} du &= -k dt \end{aligned}$$

Then, we integrate both sides:

$$\begin{aligned} \int \frac{1}{u - T_{\text{ext}}} du &= \int -k dt \\ \ln|u - T_{\text{ext}}| &= -kt + C \\ |u - T_{\text{ext}}| &= e^{-kt+C} \\ u - T_{\text{ext}} &= \pm e^{C} e^{-kt} \\ u(t) &= \pm e^{C} e^{-kt} + T_{\text{ext}} \\ u(t) &= Ke^{-kt} + T_{\text{ext}} \quad , \quad K = \pm e^{C} \end{aligned}$$

To find K, we use the initial condition $u(0) = u_0$:

$$\begin{aligned} u(0) &= Ke^{-k \cdot 0} + T_{\text{ext}} \\ u_0 &= K + T_{\text{ext}} \\ K &= u_0 - T_{\text{ext}} \end{aligned}$$

Therefore, the solution to the ODE is:

$$\boxed{u(t) = (u_0 - T_{\text{ext}})e^{-kt} + T_{\text{ext}}}$$

Intuitively, this can also be used to model warming as well as cooling. This depends on the sign of $(u_0 - T_{\text{ext}})$.

2.2 Integrating factors

We are given an equation:

$$\dot{x} + p(t)x = q(t)$$

The method of integrating factors can be thought of intuitively as "undoing" the product rule of differentiation. Specifically, we would like to transform the left-hand side of the equation into the derivative of a product of two functions $u(t)$ and $x(t)$:

$$\frac{d}{dt}[u(t)x(t)] = u(t)\dot{x}(t) + \dot{u}(t)x(t)$$

Looking at our original equation, we'd like to have some $\mu(t)$ to multiply throughout the equation such that:

$$\begin{aligned} \mu(t)\dot{x} + \mu(t)p(t)x &= u(t)\dot{x} + \dot{u}(t)x \\ &= \frac{d}{dt}[u(t)x(t)] \end{aligned}$$

This means we need to have:

$$\begin{cases} \mu(t) = u(t) \\ \mu(t)p(t) = \dot{u}(t) \end{cases}$$

Solving for $\mu(t)$, we have:

$$\begin{aligned}
\dot{u} &= up(t) \\
\frac{du}{dt} &= up(t) \\
\frac{1}{u}du &= p(t)dt \\
\int \frac{1}{u}du &= \int p(t)dt \\
\ln|u| &= \int p(t)dt + C \\
u(t) = e^{\int p(t)dt + C} &= e^{C}e^{\int p(t)dt}
\end{aligned}$$

We can ignore the constant as it will cancel out later.

Definition 2.2.1: Integrating factor

The **integrating factor** for the first-order linear ODE $\dot{x} + p(t)x = q(t)$ is given by:

$$\boxed{\mu(t) = e^{\int p(t)dt}}$$

Also, this is the reciprocal of the homogeneous solution $X_h(t)$ from separation of variables:

$$\mu(t) = X_h^{-1}(t)$$

2.2.1 Steps to solve using integrating factors

1. Start with the standard form of the ODE: $\dot{x} + p(t)x = q(t)$.
2. Compute the integrating factor: $\mu(t) = e^{\int p(t)dt}$.
3. Multiply both sides of the ODE by the integrating factor $\mu(t)$.
4. Recognize that the left-hand side is now the derivative of the product:
$$\frac{d}{dt}\left[\mu(t)x(t)\right] = \mu(t)q(t)$$
5. Integrate both sides with respect to t:
$$\mu(t)x(t) = \int \mu(t)q(t)dt + C$$
6. Solve for $x(t)$:
$$x(t) = \frac{1}{\mu(t)}\left(\int \mu(t)q(t)dt + C\right)$$
7. (Optional) Apply any initial conditions to solve for the constant C.

Example 2.2.1

Solve the initial value problem:

$$t\dot{x} + 2x = t^2 - t + 1, \quad x(1) = \frac{1}{2}$$

Solution:

1. Standard form: $\dot{x} + \frac{2}{t}x = t - 1 + \frac{1}{t}$
2. Integrating factor:

$$\begin{aligned}\mu(t) &= e^{\int \frac{2}{t}dt} \\ &= e^{2\ln|t|} \\ &= t^2\end{aligned}$$

3. Multiply through by $\mu(t)$:

$$t^2\dot{x} + 2tx = t^3 - t^2 + t$$

4. Left-hand side as derivative:

$$\frac{d}{dt}\left[t^2 x\right] = t^3 - t^2 + t$$

5. Integrate both sides:

$$\begin{aligned}t^2 x &= \int (t^3 - t^2 + t)dt + C \\ &= \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2} + C\end{aligned}$$

6. Solve for $x(t)$:

$$x(t) = \frac{1}{t^2}\left(\frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2} + C\right) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{C}{t^2}$$

7. Apply initial condition $x(1) = \frac{1}{2}$:

$$\begin{aligned}\frac{1}{2} &= \frac{1}{4} - \frac{1}{3} + \frac{1}{2} + C \\ C &= \frac{1}{12}\end{aligned}$$

$$\boxed{x(t) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{1}{12t^2}}$$

2.3 Substitution technique

Consider the differential equation:

$$\dot{y} = F(\frac{y}{x})$$

We can eliminate the x in the denominator to have a function of a single variable by substituting:

$$v = \frac{y}{x} \implies y = vx$$

Then rearranging:

$$\begin{aligned}\dot{y} &= F(v) \\ v + x\dot{v} &= F(v) \\ x\dot{v} &= F(v) - v \\ x \cdot \frac{dv}{dx} &= F(v) - v \\ \frac{dv}{dx} &= \frac{F(v) - v}{x} \\ \frac{dv}{F(v) - v} &= \frac{dx}{x} \quad \text{(separable equation)}\end{aligned}$$

2.3.1 Steps to solve using substitution technique

1. Identify the substitution: $v = \frac{y}{x}$, which implies $y = vx$.
2. Differentiate y with respect to x using the product rule:
$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$
3. Substitute $\frac{dy}{dx}$ and y into the original ODE to express it in terms of v and x.
4. Rearrange the equation to isolate $\frac{dv}{dx}$.
5. Separate the variables to obtain an equation of the form:
$$\frac{dv}{F(v) - v} = \frac{dx}{x}$$
6. Integrate both sides to find $v(x)$.
7. Substitute back to find $y(x)$ using $y = vx$.
8. (Optional) Apply any initial conditions to solve for constants of integration.

We can now solve this separable equation for $v(x)$, then substitute back to find $y(x)$.

Example 2.3.1 (Tutorial 3 problem 1)
Solve the initial value problem:
$$\begin{cases} \dot{y} = \frac{x^2+3y^2}{2xy} \\ y(-2) = 6 \end{cases}$$

Solution: We rewrite so we can make some substitution for $u = \frac{y}{x}$:

$$\begin{aligned} \dot{y} &= \frac{x^2 + 3y^2}{2xy} \\ &= \frac{x^2}{2xy} + \frac{3y^2}{2xy} \\ &= \frac{x}{2y} + \frac{3y}{2x} \\ &= \frac{1}{2 \cdot \frac{y}{x}} + \frac{3}{2} \cdot \frac{y}{x} \\ &= \frac{1}{2u} + \frac{3}{2}u \end{aligned}$$

Recall that $u = \frac{y}{x} \implies y = ux$. Then $\frac{dy}{dx} = u + x\frac{du}{dx}$. Substituting this in:

$$\begin{aligned} u + x\frac{du}{dx} &= \frac{1}{2u} + \frac{3}{2}u \\ x\frac{du}{dx} &= \frac{1}{2u} + \frac{3}{2}u - u \\ x\ du &= \left(\frac{1}{2u} + \frac{1}{2}u\right) dx \\ \frac{1}{\frac{1}{2u} + \frac{1}{2}u}\ du &= \frac{dx}{x} \end{aligned}$$

Note that this is effectively transforming the original equation into the separable form $\frac{dv}{F(v)-v} = \frac{dx}{x}$. Now we integrate by partial fractions:

$$\begin{aligned} \int \frac{1}{\frac{1}{2u} + \frac{1}{2}u}\ du &= \int \frac{dx}{x} \\ \int \frac{2u}{1+u^2}\ du &= \ln|x| + C \\ \int \frac{1}{1+u^2}\ d(u^2) &= \ln|x| + C \\ \ln|1+u^2| &= \ln|x| + C \\ 1 + u^2 &= K|x|\ , \quad K = e^C 1 + u^2 \qquad\qquad = Bx \quad B = \pm K \end{aligned}$$

Then substituting back for $u = \frac{y}{x}$:

$$\begin{aligned} 1 + \left(\frac{y}{x}\right)^2 &= Bx \\ 1 + \frac{y^2}{x^2} &= Bx \\ y^2 &= Bx^3 - x^2 \end{aligned}$$

Rearranging gives the general solution:

$$y^2 = Bx^3 - x^2$$

We can solve for y:

$$\boxed{y = \pm\sqrt{Bx^3 - x^2}}$$

Now using the initial condition $y(-2) = 6$ to solve for B:

$$\begin{aligned} 6 &= \pm\sqrt{B(-2)^3 - (-2)^2} \\ 36 &= -8B - 4 \\ 40 &= -8B \\ B &= -5 \end{aligned}$$

We must choose the positive root since $y(-2) = 6 > 0$. Thus the particular solution is:

$$\boxed{y = \sqrt{-5x^3 - x^2}}$$

The domain is $-5x^3 - x^2 \geqslant 0 \implies x^2(-5x - 1) \geqslant 0 \implies x \leqslant -\frac{1}{5}$.

2.4 Exact equations

We have solely dealt with first-order linear and separable differential equations so far. However, some ODEs will have functions of both x and y that prevent separation.

Example 2.4.1

Solve the following ODE:

$$(2xy - 9x^2) + (2y + x^2 + 1)\frac{dy}{dx} = 0$$

Solution: Now consider some function $\Psi(x, y)$ (don't worry about how we got it yet) such that:

$$\Psi(x, y) = y^2 + (x^2 + 1)y - 3x^3$$

If we compute the partial derivatives, we find:

$$\Psi_x = 2xy - 9x^2$$
$$\Psi_y = 2y + x^2 + 1$$

These expressions appear in our original equation. Using the chain rule for partial derivatives:

$$\frac{d}{dx}\Psi(x, y(x)) = \Psi_x \frac{dx}{dx} + \Psi_y \frac{dy}{dx}$$
$$\frac{d}{dx}\Psi(x, y(x)) = \Psi_x + \Psi_y \frac{dy}{dx}$$

Then we can rewrite the original ODE as:

$$\Psi_x + \Psi_y \frac{dy}{dx} = 0$$

Thus, we have:

$$\frac{d}{dx}\Psi(x, y) = 0$$

Integrating both sides with respect to x gives:

$$\Psi(x, y) = C$$

So the general solution to the ODE is:

$$\boxed{y^2 + (x^2 + 1)y - 3x^3 = C}$$

We are therefore concerned with obtaining a method to find such a function $\Psi(x, y)$ for a given ODE.

Definition 2.4.1: Exact equation

An ODE of the form:

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

is said to be an **exact equation** if there exists a function $\Psi(x, y)$ such that:

$$\Psi_x = M(x, y)$$
$$\Psi_y = N(x, y)$$

If an ODE is exact, we have:

$$\boxed{\frac{d}{dx}\Psi(x, y) = \Psi_x + \Psi_y \frac{dy}{dx} = 0}$$

Then the solution to the ODE is given implicitly by:

$$\boxed{\Psi(x, y) = C}$$

If $\Psi(x, y)$ is continuously differentiable, then we have that:

$$(\Psi_x)_y = (\Psi_y)_x$$

Then the equation is only exact if:

$$\boxed{M_y = N_x}$$

2.4.1 Steps to solve exact equations

1. Verify that the equation is exact by checking if $M_y = N_x$.
2. Integrate $M(x, y)$ with respect to x to find $\Psi(x, y)$ up to a function of y:

$$\Psi(x, y) = \int M(x, y)\, dx + h(y)$$

3. Differentiate $\Psi(x, y)$ with respect to y and set it equal to $N(x, y)$ to solve for $h(y)$:

$$\frac{\partial \Psi}{\partial y} = N(x, y)$$

4. Substitute $h(y)$ back into $\Psi(x, y)$.
5. (Optional) Apply the initial condition if given back into:

$$\Psi(x, y) = C$$

Note:-

If it is easier, you can also integrate $N(x, y)$ with respect to y first, then differentiate with respect to x to find the function of x.

Example 2.4.2

Solve the following initial value problem:

$$2xy - 9x^2 + (2y + x^2 + 1)\frac{dy}{dx} = 0, \quad y(0) = -3$$

Solution:

1. Find and verify M_y and N_x:

$$\begin{aligned} M(x, y) &= 2xy - 9x^2 \\ N(x, y) &= 2y + x^2 + 1 \\ M_y(x, y) &= 2x \\ N_x(x, y) &= 2x \end{aligned}$$

   Since $M_y = N_x$, the equation is exact.

2. Integrate $M(x, y)$ with respect to x:

$$\begin{aligned} \Psi(x, y) &= \int (2xy - 9x^2)\, dx + h(y) \\ &= x^2 y - 3x^3 + h(y) \end{aligned}$$

3. Differentiate $\Psi(x, y)$ with respect to y and set equal to $N(x, y)$:

$$\begin{aligned} \frac{\partial \Psi}{\partial y} &= x^2 + h'(y) \\ x^2 + h'(y) &= 2y + x^2 + 1 \\ h'(y) &= 2y + 1 \end{aligned}$$

   Integrating gives:

$$h(y) = y^2 + y + K$$

4. Substitute $h(y)$ back into $\Psi(x, y)$:

$$\Psi(x, y) = x^2y - 3x^3 + y^2 + y + K$$

5. Apply the initial condition $y(0) = -3$:

$$\begin{aligned}\Psi(0, -3) &= 0^2 \cdot (-3) - 3 \cdot 0^3 + (-3)^2 + (-3) + K \\ &= 0 + 0 + 9 - 3 + K \\ &= 6 + K\end{aligned}$$

Setting this equal to C, we have:

$$C = 6 + K$$

Thus, the particular solution is:

$$\boxed{x^2y - 3x^3 + y^2 + y = 6}$$

6. We could also solve for y explicitly using the quadratic formula:

$$y^2 + (x^2 + 1)y + (-3x^3 - 6) = 0$$

$$\begin{aligned}y &= \frac{-(x^2 + 1) \pm \sqrt{(x^2 + 1)^2 - 4(-3x^3 - 6)}}{2} \\ y &= \frac{-(x^2 + 1) \pm \sqrt{x^4 + 2x^2 + 1 + 12x^3 + 24}}{2} \\ y &= \frac{-(x^2 + 1) \pm \sqrt{x^4 + 12x^3 + 2x^2 + 25}}{2}\end{aligned}$$

We must now choose the correct sign for the initial condition $y(0) = -3$, since -3 is less than $\frac{-(0^2+1)}{2} = -\frac{1}{2}$. Thus we choose the negative root:

$$\boxed{y = \frac{-(x^2 + 1) - \sqrt{x^4 + 12x^3 + 2x^2 + 25}}{2}}$$

2.5 Non-exact equations

Sometimes, we will have that $M_y \neq N_x$. Like with integrating factors, we can multiply through by some function $\mu(x, y)$ to make the equation exact.

Let there be some function $\mu(x, y)$ such that:

$$\mu M + \mu N \dot{y} = 0$$

Then for exactness, we need:

$$(\mu M)_y = (\mu N)_x$$

Expanding both sides using the product rule:

$$\begin{aligned}(\mu M)_y &= \mu_y M + \mu M_y \\ (\mu N)_x &= \mu_x N + \mu N_x\end{aligned}$$

This is a partial differential equation in $\mu(x, y)$. In general, this is difficult to solve. However, if we assume that μ is only a function of x or only a function of y, we can reduce this to an ordinary differential equation.

Definition 2.5.1: Integrating factor for non-exact equations

An **integrating factor** for a non-exact equation of the form:

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

is a function $\mu(x)$ or $\mu(y)$ such that multiplying through by μ makes the equation exact.

1. **Case 1: $\mu = \mu(x)$ (function of only x).** Then if we rearrange for $\frac{d\mu}{dx}$, we have:

$$\begin{aligned}
\mu_y M + \mu M_y &= \mu_x N + \mu N_x \\
0 + \mu M_y &= \frac{d\mu}{dx} N + \mu N_x \\
\mu M_y - \mu N_x &= \frac{d\mu}{dx} N \\
\frac{1}{\mu}\frac{d\mu}{dx} &= \frac{M_y - N_x}{N} \\
\frac{d\mu}{dx} &= \mu \cdot \frac{M_y - N_x}{N}
\end{aligned}$$

As this is now separable, we can write the integrating factor as:

$$\boxed{\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}}$$

2. **Case 2: $\mu = \mu(y)$ (function of only y).** Then if we rearrange for $\frac{d\mu}{dy}$, we have:

$$\begin{aligned}
\mu_y M + \mu M_y &= \mu_x N + \mu N_x \\
\frac{d\mu}{dy} M + \mu M_y &= 0 + \mu N_x \\
\frac{d\mu}{dy} M &= \mu N_x - \mu M_y \\
\frac{1}{\mu}\frac{d\mu}{dy} &= \frac{N_x - M_y}{M} \\
\frac{d\mu}{dy} &= \mu \cdot \frac{N_x - M_y}{M}
\end{aligned}$$

Then we can write the integrating factor as:

$$\boxed{\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}}$$

Note:-

This method only works if we can find an integrating factor that is solely a function of x or solely a function of y. So we must first check if either $\frac{M_y - N_x}{N}$ is a function of x only or $\frac{N_x - M_y}{M}$ is a function of y only.

2.5.1 Steps to solve non-exact equations using integrating factors

1. Compute M_y and N_x.
2. Check if $\frac{M_y - N_x}{N}$ is a function of x only or if $\frac{N_x - M_y}{M}$ is a function of y only.
3. If $\frac{M_y - N_x}{N}$ is a function of x only, compute the integrating factor:

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$$

   If $\frac{N_x - M_y}{M}$ is a function of y only, compute the integrating factor:

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$$

4. Multiply the original ODE by the integrating factor $\mu(x)$ or $\mu(y)$.

5. Verify that the new equation is exact by checking if the new $M_y = N_x$.
6. Solve the exact equation using the steps for exact equations.
7. (Optional) Apply any initial conditions to solve for constants of integration.

Example 2.5.1

Find a general solution to the ODE:

$$y + (2x - ye^y)\frac{dy}{dx} = 0$$

Solution:

1. We compute M_y and N_x:

$$\begin{aligned} M(x,y) &= y \\ N(x,y) &= 2x - ye^y \\ M_y(x,y) &= 1 \\ N_x(x,y) &= 2 \end{aligned}$$

   Then this is not exact since $M_y \neq N_x$.

2. We check if $\frac{M_y - N_x}{N}$ is a function of x only or if $\frac{N_x - M_y}{M}$ is a function of y only:

$$\begin{aligned} \frac{M_y - N_x}{N} &= \frac{1-2}{2x - ye^y} = \frac{-1}{2x - ye^y} \quad \text{(not a function of } x \text{ only)} \\ \frac{N_x - M_y}{M} &= \frac{2-1}{y} = \frac{1}{y} \quad \text{(function of } y \text{ only)} \end{aligned}$$

3. We compute the integrating factor:

$$\begin{aligned} \mu(y) &= e^{\int \frac{1}{y} dy} \\ &= e^{\ln|y|} \\ &= |y| \end{aligned}$$

4. We multiply through by $\mu(y) = |y|$:

$$y^2 + |y|(2x - ye^y)\frac{dy}{dx} = 0$$

5. We verify exactness by computing the new M_y and N_x:

$$\begin{aligned} M(x,y) &= y^2 \\ N(x,y) &= |y|(2x - ye^y) \\ M_y(x,y) &= 2y \\ N_x(x,y) &= 2|y| \end{aligned}$$

   Since $M_y = N_x$, the new equation is exact.

6. We solve the exact equation:

$$\begin{aligned} \Psi(x,y) &= \int y^2\, dx + h(y) \\ &= xy^2 + h(y) \end{aligned}$$

Differentiating with respect to y and setting equal to $N(x, y)$:

$$\begin{aligned}
\Psi_y &= 2xy + h'(y) \\
2xy + h'(y) &= |y|(2x - ye^y) \\
h'(y) &= |y|(2x - ye^y) - 2xy \\
h'(y) &= -|y|ye^y \\
h'(y) &= -y^2e^y \quad \text{(since } y^2 \geqslant 0\text{, we can drop the absolute value)}
\end{aligned}$$

Integrating gives:

$$\begin{aligned}
h(y) &= \int -y^2e^y \, dy \\
&= -(y^2 - 2y + 2)e^y + K
\end{aligned}$$

Substituting back into $\Psi(x, y)$:

$$\Psi(x, y) = xy^2 - (y^2 - 2y + 2)e^y + K$$

Thus, the general solution is given implicitly by:

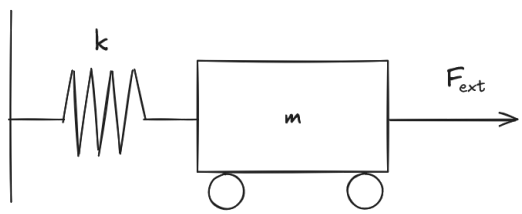
$$\boxed{xy^2 - (y^2 - 2y + 2)e^y = C}$$

Chapter 3

Techniques for solving second-order ODEs

3.1 Classical mechanics: mass-spring system

Consider a mass m attached to a spring with spring constant k. It is also subject to an external force F_{ext}.



Note that velocity is the first derivative of position with respect to time, denoted $\dot{x}(t)$, and acceleration is the second derivative of position with respect to time, denoted $\ddot{x}(t)$.

Hooke's law gives the force exerted by the spring:

$$F_{\text{spring}} = -kx, \quad k > 0$$

Then, by Newton's second law, the total force on the mass is equal to its mass times its acceleration:

$$m\ddot{x} = F_{\text{spring}} + F_{\text{ext}}$$

Rearranging gives the second-order ODE:

$$m\ddot{x} + kx = F_{\text{ext}}$$

We could also apply a damping force (such as friction) proportional to velocity, with damping coefficient b. We call this:

$$F_{\text{damping}} = -b\dot{x}, \quad b > 0$$

Then the ODE becomes:

$$m\ddot{x} + b\dot{x} + kx = F_{\text{ext}}$$

Definition 3.1.1: Linear ODE

A linear ODE is one of the form:

$$a_m x^{(m)} + a_{m-1} x^{(m-1)} + \cdots + a_1 \dot{x} + a_0 x = q(t)$$

where $a_0, a_1, \ldots, a_m$ are coefficients that may depend on t but not on x or its derivatives, and $q(t)$ is a function of t.
We call this a m-th order linear ODE.

In our mass-spring system, we will have two initial conditions.

1. Initial position: $x(0) = x_0$
2. Initial velocity: $\dot{x}(0) = v_0$

Example 3.1.1 (A special case)

Consider a mass-spring system where $b = 0$ (no damping) and there is no external force ($F_{\text{ext}} = 0$). Solve the ODE:

$$m\ddot{x} + kx = 0$$

with initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$.

Solution: Without knowing any specific methods, we rearrange the equation:

$$\ddot{x} = -\frac{k}{m}x$$

So we need some function whose second derivative is proportional to the negative of the function itself. We know that sine and cosine functions have this property.

Try $x = \sin(\omega t)$, and differentiate several times:

$$\begin{aligned}\dot{x} &= \omega\cos(\omega t)\\ \ddot{x} &= -\omega^2\sin(\omega t)\end{aligned}$$

Then substituting into the ODE gives:

$$-m\omega^2\sin(\omega t) + k\sin(\omega t) = 0$$

$$\boxed{x(t) = \sin(\omega t), \quad \text{where } \omega = \sqrt{\frac{k}{m}}}$$

This is only a particular solution (you could try with cosine as well, and the general solution is a linear combination of both). Intuitively, the mass will oscillate back and forth indefinitely.

Theorem 3.1.1 Superposition principle (applied to second-order linear ODEs)

For a linear ODE, if $x_1(t)$ and $x_2(t)$ are solutions to the homogeneous equation (i.e., when $q(t) = 0$), then any linear combination of these solutions is also a solution. That is:

$$x(t) = C_1x_1(t) + C_2x_2(t)$$

is also a solution for any constants C_1 and C_2. The functions $x_1(t)$ and $x_2(t)$ are called the **modes** of the system.

Proof: Let $x_1(t)$ and $x_2(t)$ be solutions to the homogeneous equation:

$$a_2\ddot{x} + a_1\dot{x} + a_0x = 0$$

Then, substituting the potential solution $x(t) = C_1x_1(t) + C_2x_2(t)$:

$$\begin{aligned}a_2\ddot{x} + a_1\dot{x} + a_0x &= a_2\left(C_1\ddot{x}_1 + C_2\ddot{x}_2\right) + a_1\left(C_1\dot{x}_1 + C_2\dot{x}_2\right) + a_0\left(C_1x_1 + C_2x_2\right)\\ &= C_1\left(a_2\ddot{x}_1 + a_1\dot{x}_1 + a_0x_1\right) + C_2\left(a_2\ddot{x}_2 + a_1\dot{x}_2 + a_0x_2\right)\\ &= C_1\cdot 0 + C_2\cdot 0 = 0\end{aligned}$$

Thus, $x(t) = C_1x_1(t) + C_2x_2(t)$ is also a solution. 😄

Example 3.1.2 (General solution)

Find the general solution to the ODE:

$$m\ddot{x} + kx = 0$$

Solution: From the previous example, we know that both $\sin(\omega t)$ and $\cos(\omega t)$ are solutions, where $\omega = \sqrt{\frac{k}{m}}$.

Therefore, by the superposition principle, the general solution is:

$$\boxed{x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t)}$$

where C_1 and C_2 are constants determined by the initial conditions.

Note:-

The functions $\sin(\omega t)$ and $\cos(\omega t)$ are linearly independent because to transform one into another, you would need to add some factor in the argument (i.e., phase shift). This is not a scalar multiple transformation, so no linear transformation exists between them.

3.2 Equations with constant coefficients

3.2.1 Deriving the characteristic equation

We make the ansatz (educated guess) that the solution is of the form:

$$x(t) = ce^{rt}$$

where r is a constant to be determined. This guess is motivated by the fact that exponentials have the property that their derivatives are proportional to themselves (remember how we worked with the sin and cos functions earlier).

We want to attempt to form a linear combination of such solutions.

Recalling our equation of the form:

$$m\ddot{x} + b\dot{x} + kx = 0$$

Substituting our ansatz into the ODE gives:

$$mr^2ce^{rt} + brce^{rt} + kce^{rt} = 0$$

The trivial solution is $c = 0$, but we are interested in non-trivial solutions where $c \neq 0$. Dividing both sides by ce^{rt} (which is never zero) gives the characteristic equation:

$$mr^2 + br + k = 0$$

This is the **characteristic equation** of the ODE. It can be observed that we will obtain two roots, or two **modes**. This makes sense - we expect two linearly independent solutions for a second-order ODE, with the general solution being a linear combination of these two solutions.

3.2.2 Distinct real roots

Example 3.2.1

Find the general solution to the ODE:

$$\ddot{x} + 5\dot{x} + 4x = 0$$

Then, find the specific solution satisfying the initial conditions $x(0) = 2$ and $\dot{x}(0) = -5$.

Solution: We write this as

$$1 \cdot \ddot{x} + 5 \cdot \dot{x} + 4 \cdot x = 0$$

so that we can identify $m = 1$, $b = 5$, and $k = 4$. Then, the characteristic equation is:

$$P(s) = 1 \cdot s^2 + 5 \cdot s + 4 = 0$$

Solving for the roots, we have:

$$r^2 + 5r + 4 = 0$$

Factoring gives:

$$(r + 4)(r + 1) = 0$$

Thus, the roots are $r_1 = -4$ and $r_2 = -1$.

Therefore, the general solution is:

$$\boxed{x(t) = C_1 e^{-4t} + C_2 e^{-t}}$$

where C_1 and C_2 are constants determined by initial conditions.

To find the specific solution, we apply the initial conditions:

$$\begin{aligned} x(0) &= C_1 e^0 + C_2 e^0 = C_1 + C_2 = 2 \quad (1) \\ \dot{x}(t) &= -4C_1 e^{-4t} - C_2 e^{-t} \\ \dot{x}(0) &= -4C_1 e^0 - C_2 e^0 = -4C_1 - C_2 = -5 \quad (2) \end{aligned}$$

Solving equations (1) and (2): From (1): $C_2 = 2 - C_1$. Substituting into (2):

$$\begin{aligned} -4C_1 - (2 - C_1) &= -5 \\ -4C_1 - 2 + C_1 &= -5 \\ -3C_1 - 2 &= -5 \\ -3C_1 &= -3 \\ C_1 &= 1 \end{aligned}$$

Then from (1):

$$C_2 = 2 - 1 = 1$$

Thus, the specific solution is:

$$\boxed{x(t) = e^{-4t} + e^{-t}}$$

With an equation of the form $m\ddot{x} + b\dot{x} + kx = 0$, we can summarize the types of solutions based on the discriminant $D = b^2 - 4mk$:

- **Overdamped** ($D > 0$): Two distinct real roots, leading to solutions that decay exponentially without oscillating.
$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$
- **Critically damped** ($D = 0$): One repeated real root, leading to solutions that decay to zero as quickly as possible without oscillating.
$$x(t) = C_1 e^{rt} + C_2 t e^{rt}$$
- **Underdamped** ($D < 0$): Complex conjugate roots, leading to **oscillatory** solutions that decay exponentially.

3.2.3 Review of complex numbers

Note:-

The lectures started from the introduction of $i^2 = -1$. These notes skip ahead to the parts more relevant to ODEs.

Definition 3.2.1: Argument of a complex number

The **argument** of a complex number $z = a + bi$ is the angle θ formed with the positive real axis in the complex plane. It is denoted as $\arg(z)$.

The argument can be calculated using the arctangent function:

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

Some useful properties of complex numbers and their arguments:

$$|z_1 z_2| = |z_1||z_2|$$
$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$
$$\arg(z^n) = n \arg(z)$$

We can parametrize a complex number on the unit circle by letting $t = \arg(z)$:

$$z = \cos t + i \sin t$$

Then, differentiating:

$$\dot{z} = -\sin t + i \cos t$$

This then implies a useful identity:

$$\boxed{\dot{z} = iz}$$

Solving this ODE gives:

$$z(t) = z(0)e^{it}$$

Euler's identity states that:

$$\boxed{e^{it} = \cos t + i \sin t}$$

This is useful because it allows us to express trigonometric functions in terms of exponentials.
Another interesting identity arises when we set $t = \pi$:

$$e^{i\pi} + 1 = 0$$

Now, consider a complex number $z = a + bi$. We have that:

$$\begin{aligned} \dot{z} &= (a + bi)z \\ z(t) &= z(0)e^{(a+bi)t} \\ &= z(0)e^{at}(\cos bt + i \sin bt) \end{aligned}$$

where e^{at} represents the magnitude (scaling) and $\cos bt + i \sin bt$ represents the rotation in the complex plane (argument).
Another useful property of complex numbers:

$$\boxed{\mathrm{Re}(z) = \frac{1}{2}(z + \overline{z})}$$

$$\boxed{\mathrm{Im}(z) = \frac{1}{2i}(z - \overline{z})}$$

where $\overline{z} = a - bi$ is the complex conjugate of $z = a + bi$.
Then we define cos and sin in terms of exponentials:

$$\boxed{\cos t = \frac{1}{2}\left(e^{it} + e^{-it}\right)}$$

$$\boxed{\sin t = \frac{1}{2i}\left(e^{it} - e^{-it}\right)}$$

3.2.4 Complex roots

Theorem 3.2.1

If z is a complex-valued solution to $m\ddot{z} + b\dot{z} + kz = 0$, then both the real part $\mathrm{Re}(z)$ and the imaginary part

Im(z) are also solutions to the ODE.

Proof: This should not be surprising. If we write $z = u(t) + iv(t)$ where $u(t) = \text{Re}(z)$ and $v(t) = \text{Im}(z)$, then substituting into the ODE gives:

$$\begin{aligned} m\ddot{z} + b\dot{z} + kz &= m(\ddot{u} + i\ddot{v}) + b(\dot{u} + i\dot{v}) + k(u + iv) \\ &= (m\ddot{u} + b\dot{u} + ku) + i(m\ddot{v} + b\dot{v} + kv) \end{aligned}$$

We basically get two separate equations, corresponding to the real and imaginary parts:

$$\begin{aligned} m\ddot{u} + b\dot{u} + ku &= 0 \\ m\ddot{v} + b\dot{v} + kv &= 0 \end{aligned}$$

Since z is a solution, both parts must equal zero. Thus, both Re(z) and Im(z) are solutions to the ODE. 😄

We can observe that an oscillatory solution arises when the roots of the characteristic equation are complex conjugates. To derive this, we use the second-order ODE:

$$r_1 = a + bi, \quad r_2 = a - bi$$

Extracting the real and imaginary parts gives us two linearly independent solutions:

$$\begin{aligned} x_1(t) &= e^{(a+bi)t} = e^{at}\cos(bt) \\ x_2(t) &= e^{(a-bi)t} = e^{at}\sin(bt) \end{aligned}$$

So the general solution is a linear combination of these two solutions (where a determines the exponential decay or growth - the real part, and b determines the frequency of oscillation - the imaginary part):

$$\boxed{x(t) = C_1 e^{at}\cos(bt) + C_2 e^{at}\sin(bt)}$$

We can also represent this as a phase shifted cosine function:

$$\boxed{x(t) = Ae^{at}\cos(bt - \phi)}$$

where A is the amplitude and ϕ is the phase shift, which can be determined from the constants C_1 and C_2. Formulas for A and ϕ are:

$$\boxed{A = \sqrt{C_1^2 + C_2^2}}$$

$$\boxed{\phi = \tan^{-1}\left(\frac{C_2}{C_1}\right)}$$

The identity used to get this phase shift:

$$\begin{aligned} \cos(\alpha - \beta) &= \cos\alpha\cos\beta + \sin\alpha\sin\beta \\ &= \frac{C_1}{A}\cos bt + \frac{C_2}{A}\sin bt \end{aligned}$$

Example 3.2.2

Find the general solution to the ODE:

$$\ddot{x} + 4\dot{x} + 5x = 0$$

Solution: We write this as

$$1 \cdot \ddot{x} + 4 \cdot \dot{x} + 5 \cdot x = 0$$

so that we can identify $m = 1$, $b = 4$, and $k = 5$. Then, the characteristic equation is:

$$P(s) = 1 \cdot s^2 + 4 \cdot s + 5 = 0$$

Solving for the roots, we have:

$$r^2 + 4r + 5 = 0$$

Using the quadratic formula:

$$r = \frac{-4 \pm \sqrt{16 - 20}}{2} = -2 \pm i$$

Thus, the roots are $r_1 = -2 + i$ and $r_2 = -2 - i$. Therefore, the general solution is:

$$\boxed{x(t) = e^{-2t}(C_1 \cos t + C_2 \sin t)}$$

We can also express this as a phase shifted cosine function:

$$\boxed{x(t) = Ae^{-2t} \cos(t - \phi)}$$

3.2.5 Repeated roots

Let $P(D)$ be the polynomial differential operator associated with a linear ODE, where Q is a polynomial with no repeated roots. If r is a root of multiplicity m of the characteristic polynomial $P(\lambda)$, then

$$P(\lambda) = (\lambda - r)^m Q(\lambda) \quad \Rightarrow \quad P(D) = (D - r)^m Q(D).$$

Since $Q(r) \neq 0$:

$$(D - r)^m y = 0.$$

Solving successively,

$$\begin{aligned} (D - r)y &= 0 \\ Dy - ry &= 0 \\ \frac{dy}{dt} &= ry \\ y &= e^{rt} \end{aligned}$$

and each additional application of $(D - r)^{-1}$ introduces a factor of t. Hence the m linearly independent solutions are

$$e^{rt},\ te^{rt},\ t^2e^{rt},\ \ldots,\ t^{m-1}e^{rt}.$$

Example 3.2.3

Find the general solution to the ODE:

$$\ddot{x} - 4\dot{x} + 4x = 0$$

Solution: We write this as

$$1 \cdot \ddot{x} - 4 \cdot \dot{x} + 4 \cdot x = 0$$

so that we can identify $m = 1$, $b = -4$, and $k = 4$. Then, the characteristic equation is:

$$P(s) = 1 \cdot s^2 - 4 \cdot s + 4 = 0$$

Solving for the roots, we have:

$$r^2 - 4r + 4 = 0$$

Factoring gives:

$$(r - 2)^2 = 0$$

Thus, there is a repeated root at $r = 2$.

Therefore, the general solution is:

$$\boxed{x(t) = C_1e^{2t} + C_2te^{2t}}$$

3.3 Non-homogeneous equations

We were previous using $m\ddot{x} + b\ddot{x} + kx = 0$ to represent a situation where a mass is attached to a spring with no external damping force.

If we introduce an external force F_{ext}, then the ODE becomes:

$$m\ddot{x} + b\dot{x} + kx = F_{\text{ext}}$$

We know the solution to the homogeneous equation. Also, if $b = 0$, we can define the **natural frequency** as:

$$w_m = \sqrt{\frac{k}{m}}$$

such that the solution is:

$x(t) = C_1 \cos(w_m t) + C_2 \sin(w_m t)$ from the characteristic equation solved via the quadratic formula

And if $b \neq 0$, we can define the **damped frequency** as:

$$w_d = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad \text{from the characteristic equation solved via the quadratic formula}$$

such that the solution is:

$$x(t) = e^{-\frac{b}{2m}t}(C_1 \cos(w_d t) + C_2 \sin(w_d t))$$

We standardize the homogeneous solution as $x_h(t)$:

$$\ddot{x} + \frac{b}{m}\dot{x} + w_m^2 x = 0$$

where w_m is the natural frequency.

Theorem 3.3.1

If $x_p(t)$ is a particular solution to the non-homogeneous equation $m\ddot{x} + b\dot{x} + kx = F_{\text{ext}}$, then the general solution is given by:

$$x(t) = x_h(t) + x_p(t)$$

where $x_h(t)$ is the general solution to the corresponding homogeneous equation.

Proof: Let $x_h(t)$ be the general solution to the homogeneous equation:

$$m\ddot{x} + b\dot{x} + kx = 0$$

Then, we can write the general solution to the non-homogeneous equation as:

$$x(t) = x_h(t) + x_p(t)$$

Substituting into the non-homogeneous equation gives:

$$\begin{aligned} m\ddot{x} + b\dot{x} + kx &= m(\ddot{x}_h + \ddot{x}_p) + b(\dot{x}_h + \dot{x}_p) + k(x_h + x_p) \\ &= (m\ddot{x}_h + b\dot{x}_h + kx_h) + (m\ddot{x}_p + b\dot{x}_p + kx_p) \\ &= 0 + F_{\text{ext}} = F_{\text{ext}} \end{aligned}$$

Thus, $x(t) = x_h(t) + x_p(t)$ is indeed a solution to the non-homogeneous equation. Since $x_h(t)$ represents the general solution to the homogeneous equation, this form captures all possible solutions to the non-homogeneous equation. 😄

3.3.1 Steps to solve a non-homogeneous linear ODE

1. Find the homogeneous solution $x_h(t)$ by solving the characteristic equation.
2. Find a particular solution $x_p(t)$ using an appropriate method (e.g., undetermined coefficients, variation of parameters).
3. Combine the homogeneous and particular solutions to get the general solution:

$$x(t) = x_h(t) + x_p(t)$$

4. (Optional) Apply initial conditions to find the specific solution if needed.

3.3.2 Method of undetermined coefficients

Suppose that we have a non-homogeneous ODE of the form:

$$m\ddot{x} + b\dot{x} + kx = F_{\text{ext}}(t)$$

We suppose that $F_{\text{ext}}(t) = ky(t)$ where $y(t)$ is a known function. We also consider the case with no damping first ($b = 0$):

$$m\ddot{x} + kx = ky(t)$$

Rearranging gives:

$$m\ddot{x} + k(x - y) = 0$$

We think of $x - y$ as the deviation from the equilibrium position y. From a physical perspective, the system will oscillate around the equilibrium position y, which sets some natural frequency of oscillation.

$$y(t) = A\cos(\omega t)$$

Then inputting wih $\omega_m = \sqrt{\frac{k}{m}}$ gives:

$$\ddot{x} + \omega_m^2 x = \omega_m^2 A\cos(\omega t)$$

Seeing the cos term, we guess a particular solution of the form:

$$x_p(t) = B\cos(\omega t)$$

Plugging this into the ODE gives:

$$\begin{aligned} x_p(t) &= B\cos(\omega t) \\ \ddot{x}_p &= -B\omega^2\cos(\omega t) \\ -B\omega^2\cos(\omega t) + \omega_m^2 B\cos(\omega t) &= \omega_m^2 A\cos(\omega t) \end{aligned}$$

This is true if:

$$\boxed{B = \frac{\omega_m^2}{\omega_m^2 - \omega^2}A}$$

where ω_m is the natural frequency of the system and ω is the frequency of the external force. We can see that if ω is close to ω_m, then B becomes very large, which corresponds to the phenomenon of **resonance**. In reality, with $\omega_m = \omega$, we would need to modify our guess. We will later see how to deal with resonance.

In general, if $F_{\text{ext}}(t) = A\cos(\omega t) + B\sin(\omega t)$ appears to be a sinusoidal function, we can guess:

$$x_p(t) = C\cos(\omega t) + D\sin(\omega t)$$

Then, we can plug this into the ODE and solve for C and D.

3.3.3 Exponential inputs

Similarly, we can have $F_{\text{ext}}(t) = Ae^{rt}$, which gives:

$$m\ddot{x} + b\dot{x} + kx = Ae^{rt}$$

Then, we can guess a particular solution of the form:

$$x_p(t) = Be^{rt}$$

Plugging this into the ODE gives:

$$\begin{aligned} x_p(t) &= Be^{rt} \\ \dot{x}_p &= rBe^{rt} \\ \ddot{x}_p &= r^2Be^{rt} \\ mr^2Be^{rt} + brBe^{rt} + kBe^{rt} &= Ae^{rt} \end{aligned}$$

So we can see that to solve for B, we can divide both sides by e^{rt} (which is never zero) to get:

$$B = \frac{A}{mr^2 + br + k}$$

Note:-

In general, when we have an mth order ODE, we can use a similar approach to find particular solutions for exponential inputs. We have:

$$a_m x^{(m)} + a_{m-1}x^{(m-1)} + \cdots + a_1\dot{x} + a_0 x = Ae^{rt}$$

Then, we apply the function:

$$P(r) = a_m r^m + a_{m-1}r^{m-1} + \cdots + a_1 r + a_0$$

We have thus obtained the **exponential response formula (ERF)**:

$$\boxed{x_p(t) = \frac{Ae^{rt}}{p(r)} \quad \text{where } p(r) = a_m r^m + a_{m-1}r^{m-1} + \cdots + a_1 r + a_0 \quad p(r) \neq 0}$$

Example 3.3.1

Find the particular solution of:

$$\ddot{x} + \dot{x} + 2x = 4e^{3t}$$

Then find the general solution.

Solution: We can identify the homogeneous part of the ODE as:

$$\ddot{x} + \dot{x} + 2x = 0$$

The characteristic equation is:

$$r^2 + r + 2 = 0$$

Solving for the roots gives:

$$r = \frac{-1 \pm \sqrt{1-8}}{2} = -\frac{1}{2} \pm i\frac{\sqrt{7}}{2}$$

Thus, the homogeneous solution is:

$$x_h(t) = e^{-\frac{1}{2}t}\left(C_1 \cos\left(\frac{\sqrt{7}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{7}}{2}t\right)\right)$$

For the particular solution, we can apply the exponential response formula:

$$p(r) = r^2 + r + 2$$

Then, evaluating at $r = 3$ gives:

$$p(3) = 3^2 + 3 + 2 = 14$$

Thus, the particular solution is:

$$x_p(t) = \frac{4e^{3t}}{14} = \frac{2}{7}e^{3t}$$

Finally, the general solution is:

$$\boxed{x(t) = e^{-\frac{1}{2}t}\left(C_1 \cos\left(\frac{\sqrt{7}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{7}}{2}t\right)\right) + \frac{2}{7}e^{3t}}$$

Example 3.3.2

Find the particular solution of:

$$\ddot{x} + \dot{x} + 2x = \cos t$$

Solution: We first complexify the equation as:

$$\ddot{z} + \dot{z} + 2z = e^{it}$$

It is possible to do this because the real and imaginary parts of the solution will also be solutions to the original equation, so we can extract the real part at the end to get the particular solution.

$$p(s) = s^2 + s + 2$$

Then evaluating at $s = i$ gives:

$$p(i) = i^2 + i + 2 = 1 + i$$

Applying ERF gives:

$$z_p(t) = \frac{e^{it}}{1+i} = \frac{1-i}{2}e^{it} = \frac{1}{2}(1-i)e^{it}$$

Thus the particular solution to this complexified equation is:

$$\begin{aligned} z_p(t) &= \frac{1}{2}e^{it} - \frac{i}{2}e^{it} \\ &= \frac{1}{2}(\cos t + i \sin t) - \frac{i}{2}(\cos t + i \sin t) \\ &= \frac{1}{2}\cos t + \frac{i}{2}\sin t - \frac{i}{2}\cos t + \frac{1}{2}\sin t \\ &= \frac{1}{2}(\cos t + \sin t) + i\left(\frac{1}{2}(\sin t - \cos t)\right) \end{aligned}$$

We can then extract the real part to get the particular solution to the original equation:

$$\begin{aligned} x_p(t) &= \mathrm{Re}(z_p(t)) \\ &= \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ &= \frac{\sqrt{2}}{2}\cos\left(t - \frac{\pi}{4}\right) \end{aligned}$$

3.3.4 Resonance

An operator takes one function as input and produces another function as output. For example, the operator D takes a function $f(t)$ and produces its derivative $\dot{f}(t)$. We can also have higher-order operators like D^2 which takes a function and produces its second derivative.

The **identity operator** I is:

$$Ix = x$$

We can take linear combinations of operators. For example:

$$(D^2 + 2D + 2I)x = \ddot{x} + 2\dot{x} + 2x$$

The characteristic polynomial associated with this operator is:

$$p(s) = s^2 + 2s + 2$$

Meanwhile, $P(D)$ represents the linear combination of operators:

$$p(D) = D^2 + 2D + 2I$$

Then, we can write the ODE as:

$$p(D)x$$

Now consider an input function $q(t)$, which gives us the non-homogeneous ODE:

$$a_n x^{(n)} + a_{n-1}x^{(n-1)} + \cdots + a_1\dot{x} + a_0 x = q(t)$$

We extract the characteristic polynomial:

$$p(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$$

Then, we can write the ODE as:

$$p(D)x = q(t)$$

So the operator $p(D)$ represents the system with $x(t)$ as the response, and the input is represented by $q(t)$. Let $Lx = q$ for some operator L. Our goal is to find the inverse operator L^{-1} such that:

$$x = L^{-1}q$$

Recall the ERF which solves $p(D)x = Ae^{rt}$ for two conditions:

1. $p(r) \neq 0$.
2. A constant.

We note that if on the LHS we found the characteristic polynomial $p(s)$ with roots $r_1, r_2, \ldots, r_n$, but on the RHS we have $q(t) = Ae^{rt}$ where r is one of the roots of $p(s)$, then we have a problem because $p(r) = 0$, which means that the ERF does not apply. This corresponds to the phenomenon of **resonance**.

So we have that $p(D)x = Ae^{rt}$ where $p(r) = 0$. We'd like to make the function $p(D)$ more explicit to find some way to solve for a particular solution.

We repeatedly apply D to e^{rt}:

$$\begin{aligned} Ie^{rt} &= e^{rt} \\ De^{rt} &= re^{rt} \\ D^2e^{rt} &= r^2e^{rt} \\ &\vdots \\ D^ne^{rt} &= r^ne^{rt} \\ p(D)e^{rt} &= p(r)e^{rt} = 0 \end{aligned}$$

By the ERF, $xp(t) = \frac{1}{p(r)}e^{rt}$ is a particular solution. But $P(D)e^{rt} = p(r)e^{rt}$ is true even if $p(r) = 0$.

We start from the key identity:

$$p(D)e^{rt} = p(r)e^{rt}$$

Then we differentiate both sides with respect to r:

$$\frac{\partial}{\partial r}\left[p(D)e^{rt}\right] = \frac{\partial}{\partial r}\left[p(r)e^{rt}\right]$$

The operator $p(D)$ does not depend on r, so we can pull it out of the derivative:

$$p(D)\frac{\partial}{\partial r}e^{rt} = p'(r)e^{rt} + p(r)te^{rt}$$

Since $p(r) = 0$, this simplifies to:

$$p(D)\frac{\partial}{\partial r}e^{rt} = p'(r)e^{rt}$$

Now we get the partial derivative of e^{rt} with respect to r:

$$\frac{\partial}{\partial r}e^{rt} = te^{rt}$$

Thus, we have:

$$p(D)(te^{rt}) = p'(r)e^{rt}$$

Comparing this to the original equation $p(D)x = Ae^{rt}$, we can see that if we let $x = \frac{A}{p'(r)}te^{rt}$, then we have:

$$p(D)\left(\frac{A}{p'(r)}te^{rt}\right) = Ae^{rt}$$

Thus, we now have a formula for the particular solution in the case of resonance:

$$\boxed{x_p(t) = \frac{A}{p'(r)}te^{rt} \quad \text{where } p(r) = 0}$$

3.3.5 Generalization of resonance to higher-order ODEs

We might have a case where $p(r) = 0$, but also $p'(r) = 0$. In this case, we can differentiate the key identity again:

$$\frac{\partial^2}{\partial r^2}\left[p(D)e^{rt}\right] = \frac{\partial^2}{\partial r^2}\left[p(r)e^{rt}\right]$$

This gives us:

$$p(D)\frac{\partial^2}{\partial r^2}e^{rt} = p''(r)e^{rt} + 2p'(r)te^{rt} + p(r)t^2e^{rt}$$

Since $p(r) = 0$ and $p'(r) = 0$, this simplifies to:

$$p(D)\frac{\partial^2}{\partial r^2}e^{rt} = p''(r)e^{rt}$$

Now we get the second partial derivative of e^{rt} with respect to r:

$$\frac{\partial^2}{\partial r^2}e^{rt} = t^2e^{rt}$$

Thus, we have:

$$p(D)(t^2e^{rt}) = p''(r)e^{rt}$$

In general, the **generalized exponential response formula** for the case of resonance is:

$$\boxed{x_p(t) = \frac{A}{p^{(m+1)}(r)}t^{m+1}e^{rt} \quad \text{where } p(r) = p'(r) = \cdots = p^{(m)}(r) = 0}$$

Example 3.3.3

Find the particular solution of:

$$\ddot{x} - 4x = e^{-2t}$$

Solution: We first find the homogeneous solution by solving the characteristic equation:

$$r^2 - 4 = 0$$

which gives us the roots $r = 2$ and $r = -2$. Thus, the homogeneous solution is:

$$x_h(t) = C_1 e^{2t} + C_2 e^{-2t}$$

We see that -2 is a root of the characteristic polynomial, so we have resonance. We can apply the generalized exponential response formula to find the particular solution:

$$p(r) = r^2 - 4$$

Then we compute $p'(-2)$:

$$p'(r) = 2r \quad \Rightarrow \quad p'(-2) = -4$$

Thus, the particular solution is:

$$x_p(t) = \frac{1}{p'(-2)} t e^{-2t} = -\frac{1}{4} t e^{-2t}$$

Note:-

From a physical perspective, although we are at resonance, the particular solution does decay as t increases, which is a consequence of the fact that the input function e^{-2t} also decays as t increases. If we had an input function that did not decay, then we would have a particular solution that grows without bound as t increases, which is the classical notion of resonance.

Example 3.3.4

Find the particular solution of:

$$\ddot{x} + 4x = \cos(2t)$$

Solution: We first find the homogeneous solution by solving the characteristic equation:

$$r^2 + 4 = 0$$

which gives us the roots $r = 2i$ and $r = -2i$. Thus, the homogeneous solution is:

$$x_h(t) = C_1 \cos(2t) + C_2 \sin(2t)$$

Complexifying the equation gives us:

$$\ddot{z} + 4z = e^{2it}$$

And hence we see that $2i$ is a root of the characteristic polynomial, so we have resonance. We can apply the generalized exponential response formula to find the particular solution:

$$p(r) = r^2 + 4$$

Then we compute $p'(2i)$:

$$p'(r) = 2r \quad \Rightarrow \quad p'(2i) = 4i$$

Thus, the particular solution to the complexified equation is:

$$\begin{aligned} z_p(t) &= \frac{te^{2it}}{4i} \\ &= \frac{t}{4i}(\cos(2t) + i\sin(2t)) \\ &= \frac{t}{4}\sin(2t) - \frac{t}{4} i \cos(2t) \end{aligned}$$

Then we can extract the real part to get the particular solution to the original equation:

$$x_p(t) = \frac{t}{4}\sin(2t)$$

This solution grows without bound as t increases, which is the classical notion of resonance.

3.3.6 Steps to using complexification to solve for particular solutions

1. Complexify the equation by replacing the real input function with a complex exponential function. For example, if the input is $\cos(\omega t)$, we can replace it with $e^{i\omega t}$.
2. Apply the exponential response formula to find a particular solution to the complexified equation.
3. Extract the real part of the particular solution to get a particular solution to the original equation (if the original input was a cosine function) or extract the imaginary part to get a particular solution to the original equation (if the original input was a sine function).

3.3.7 Polynomial inputs

If we have a polynomial input, we can guess a particular solution that is also a polynomial of the same degree. For example, if we have:

$$\ddot{x} + 4x = t^2$$

Then we can guess a particular solution of the form:

$$x_p(t) = At^2 + Bt + C$$

Then we can compute $\dot{x}_p$ and $\ddot{x}_p$:

$$\begin{aligned} \dot{x}_p &= 2At + B \\ \ddot{x}_p &= 2A \end{aligned}$$

Then we can plug this into the ODE to get:

$$\begin{aligned} 2A + 4(At^2 + Bt + C) &= t^2 \\ (4A)t^2 + (4B)t + (2A + 4C) &= t^2 \end{aligned}$$

Thus, we have the system of equations:

$$\begin{aligned} 4A &= 1 \\ 4B &= 0 \\ 2A + 4C &= 0 \end{aligned}$$

which gives us $A = \frac{1}{4}$, $B = 0$, and $C = -\frac{1}{8}$. Thus, the particular solution is:

$$\boxed{x_p(t) = \frac{1}{4}t^2 - \frac{1}{8}}$$

Theorem 3.3.2 Undetermined coefficients

Given:

$$p(D)x(t) = q(t)$$

where $q(t)$ is a k-th degree polynomial, and that $p(0) \neq 0$, then there exists exactly one solution which is a polynomial of degree less than or equal to k.

Note that $p(0) = 0$ implies that $Ia_0 = 0$, or the ODE only involves derivatives. If this is the case, then this theorem may not apply.

3.3.8 Reduction of order

Second-order ODEs may also have the form:

$$p(t)\ddot{x} + q(t)\dot{x} + r(t)x = 0$$

Example 3.3.5

Find the general solution to the ODE:

$$\ddot{x} + \dot{x} = t$$

Solution: We first find the homogeneous solution by solving the characteristic equation:

$$\begin{aligned} p(s) = s^2 + s &= 0 \\ s(s+1) &= 0 \end{aligned}$$

which gives us the roots $r = 0$ and $r = -1$. Thus, the homogeneous solution is:

$$x_h(t) = C_1 + C_2 e^{-t}$$

We can use a substitution $u = \dot{x}$ to reduce the order of the ODE. Then we have $\dot{u} = \ddot{x}$, so we can rewrite the ODE as:

$$\dot{u} + u = t$$

Then if we guess $u = At + B$, and $\dot{u} = A$, we can plug this into the ODE to get:

$$\begin{aligned} A + At + B &= t \\ A &= 1 \\ B &= -1 \end{aligned}$$

Thus, we have $u(t) = t - 1$, and so $x(t) = \int u(t)dt = \int (t-1)dt = \frac{t^2}{2} - t + C_3$. We just let $C_3 = 0$ since it is already captured by the homogeneous solution. Thus, the particular solution is:

$$x_p(t) = \frac{t^2}{2} - t$$

Finally, the general solution is:

$$\boxed{x(t) = C_1 + C_2 e^{-t} + \frac{t^2}{2} - t}$$

3.3.9 Variation of parameters

We might obtain an equation like:

$$3\ddot{x} + 8\dot{x} + 6x = (t^2 + 1)e^{-t}$$

Let us suppose some function $u(t)$ exists such that $x_p(t) = u(t)e^{-t}$ is a particular solution to the above equation. Then we can compute $\dot{x}_p$ and $\ddot{x}_p$:

$$\begin{aligned} \dot{x}_p &= \dot{u}e^{-t} - ue^{-t} \\ \ddot{x}_p &= \ddot{u}e^{-t} - 2\dot{u}e^{-t} + ue^{-t} \end{aligned}$$

Then we can plug this into the ODE to get:

$$\begin{aligned} 3(\ddot{u}e^{-t} - 2\dot{u}e^{-t} + ue^{-t}) + 8(\dot{u}e^{-t} - ue^{-t}) + 6(ue^{-t}) &= (t^2+1)e^{-t} \\ 3\ddot{u}e^{-t} + 2\dot{u}e^{-t} + ue^{-t} &= (t^2+1)e^{-t} \end{aligned}$$

Then we have:

$$3\ddot{u} + 2\dot{u} + u = t^2 + 1$$

To solve this, we guess $u = At^2 + Bt + C$, and then we can compute $\dot{u}$ and $\ddot{u}$:

$$\begin{aligned} \dot{u} &= 2At + B \\ \ddot{u} &= 2A \end{aligned}$$

Then we can plug this into the ODE to get:

$$\begin{aligned} 3(2A) + 2(2At + B) + (At^2 + Bt + C) &= t^2 + 1 \\ 6A + 4At + 2B + At^2 + Bt + C &= t^2 + 1 \end{aligned}$$

Grouping by powers of t gives:

$$At^2 + (4A + B)t + (6A + 2B + C) = t^2 + 1$$

Thus we have the system of equations:

$$\begin{aligned} A &= 1 \\ 4A + B &= 0 \\ 6A + 2B + C &= 1 \end{aligned}$$

We get that $A = 1$, $B = -4$, and $C = 3$.
So we have $u(t) = t^2 - 4t + 3$, and thus the particular solution is:

$$x_p(t) = u(t)e^{-t} = (t^2 - 4t + 3)e^{-t}$$

$$\boxed{x_p(t) = (t^2 - 4t + 3)e^{-t}}$$

Note:-

We often end up with a polynomial multiplied by an exponential. To evaluate an integral involving this expression more easily, you can apply the operator trick.

$$\boxed{\int P(t)e^{\alpha t}dt = e^{\alpha t}\left(\frac{P(t)}{\alpha} - \frac{P'(t)}{\alpha^2} + \frac{P''(t)}{\alpha^3} - \cdots\right)}$$

This works because when we do integration by parts, we continue until we get a derivative equal to zero, which just returns a constant. We can also apply this operator trick to solve ODEs. For example, if we have:

$$(D - \alpha I)x = P(t)e^{\alpha t}$$

Then we can apply the operator trick to get:

$$x = \frac{P(t)}{\alpha}e^{\alpha t} - \frac{P'(t)}{\alpha^2}e^{\alpha t} + \frac{P''(t)}{\alpha^3}e^{\alpha t} - \cdots$$

Note that this is similar to how we got the generalized exponential response formula for the case of resonance, where we had to take derivatives of the characteristic polynomial until we got a non-zero value.

Chapter 4

Fourier series

Fourier series are a way to represent a periodic function as an infinite sum of sine and cosine functions. A function f is said to be **periodic** with period $2L$ if it satisfies the following condition:

$$f(t+2L) = f(t), \quad \forall t \in \mathbb{R}$$

The timepoints at which a value reoccurs $4L, 8L \ldots$ are also called **harmonics**.

Any window of width $2L$ will define the function for all t. We build up to this interval from the cosine and sine functions, which are defined as their behaviour over the window $[-\pi, \pi]$.

Note:-

$\cos(mt)$ has a period of $\frac{2\pi}{m}$, and $\sin(mt)$ has a period of $\frac{2\pi}{m}$. Furthermore, if $f(t)$ and $g(t)$ are periodic functions with period $2L$, then the function $h(t) = af(t) + bg(t)$ is also periodic with period $2L$ for any constants a and b. This is because:

$$\begin{aligned} h(t+2L) &= af(t+2L) + bg(t+2L) \\ &= af(t) + bg(t) \\ &= h(t) \end{aligned}$$

4.1 Fourier coefficients

A Fourier series has the form:

$$\boxed{f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \left(a_m \cos(mt) + b_m \sin(mt)\right)}$$

The coefficients a_0, a_m, b_m are called the **Fourier coefficients** of the function f.

Intuitively, for any given point $f(t*)$, we can think of the Fourier series as a sum of infinitely many sine and cosine waves that add up to $f(t*)$. The Fourier coefficients determine the amplitude of each sine and cosine wave in the series. The more terms we include in the series, the closer the approximation will be to the original function $f(t)$.

We wish to define the Fourier coefficients so that we can reconstruct the original function.

We begin by considering a_0. We define the function:

$$\text{Average}(f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)dt$$

Effectively, this is the average value of the function f over one period. We can also write this as:

$$\text{Average}(f) = \frac{1}{2L} \int_{-L}^{L} f(t)dt$$

The average value of $\cos(mt)$ and $\sin(mt)$ over one period is zero, so we have:

$$\text{Average}(f) = \frac{a_0}{2}$$

Thus, we can define a_0 as:

$$\boxed{a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)dt}$$

4.1.1 Properties of sin and cos integrals

The following can be obtained by using trig identities, converting to complex exponentials, or integration by parts.

$$\int_{-\pi}^{\pi} \cos(mt)\sin(mt)dt = 0$$

$$\int_{-\pi}^{\pi} \cos(mt)\cos(nt)dt = \begin{cases} 0 & m \neq n \\ \pi & m = n \neq 0 \\ 2\pi & m = n = 0 \end{cases}$$

$$\int_{-\pi}^{\pi} \sin(mt)\sin(nt)dt = \begin{cases} 0 & m \neq n \\ \pi & m = n \neq 0 \\ 0 & m = n = 0 \end{cases}$$

In general, we can compute the integrals by also considering odd and even functions. For example, $\cos(mt)\sin(nt)$ is an odd function, so its integral over $[-\pi, \pi]$ is zero.

4.1.2 Deriving the Fourier coefficients

Intuitively, a_m and b_m tell us "how much" of that term is present in the function.

To derive a_m, we use the "average" function again, and integrate over the period:

$$a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(mt)dt$$

The multiplication by $\cos(mt)$ allows us to "pick out" the a_m term from the Fourier series, since the integrals of $\cos(mt)\cos(nt)$ and $\sin(mt)\cos(nt)$ are zero for $m \neq n$. This reflects the orthogonality of the sine and cosine functions.

$$\int_{-\pi}^{\pi} a_m \cos(mt)\cos(nt)dt = \begin{cases} 0 & m \neq n \\ \pi a_m & m = n \neq 0 \\ 2\pi a_0 & m = n = 0 \end{cases}$$

Similarly, we can derive b_m as:

$$b_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(mt)dt$$

$$\int_{\pi}^{\pi} b_m \sin(mt)\sin(nt)dt = \begin{cases} 0 & m \neq n \\ \pi b_m & m = n \neq 0 \\ 0 & m = n = 0 \end{cases}$$

Then the **only** nonzero terms occur when $m = n \neq 0$.

$$\boxed{a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(mt)dt, \quad b_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(mt)dt}$$

By change of variables (on t), we can also write the Fourier coefficients as:

$$\boxed{a_m = \frac{1}{L}\int_{-L}^{L} f(t)\cos\left(\frac{m\pi t}{L}\right)dt, \quad b_m = \frac{1}{L}\int_{-L}^{L} f(t)\sin\left(\frac{m\pi t}{L}\right)dt}$$

This allows us to have the function defined on any interval of length $2L$, not just $[-\pi, \pi]$.

4.1.3 Functions defined as Fourier series

Let us consider the **square wave** function defined as:

$$f(t) = \begin{cases} 1 & 0 < t < \pi \\ -1 & -\pi < t < 0 \end{cases}$$

Notice that this function is odd.
We can compute the Fourier coefficients, starting with the constant a_0 term:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)dt = 0$$

Then, since f is odd, we have $a_m = 0$ for all m. We can compute b_m as follows:

$$\begin{aligned} b_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(mt)dt \\ &= 2 \int_{0}^{\pi} f(t) \sin(mt)dt \\ &= \frac{2}{\pi} \int_{0}^{\pi} \sin(mt)dt \\ &= \frac{2}{\pi} \left[-\frac{\cos(mt)}{m} \right]_0^{\pi} \\ &= \frac{2}{m\pi} \left[-\cos(m\pi) + 1 \right] \\ &= \frac{2}{m\pi} \left[1 - \cos(m\pi) \right] \end{aligned}$$

As a table:

m	$\cos(m\pi)$	$1 - \cos(m\pi)$
1	-1	2
2	1	0
3	-1	2
4	1	0
5	-1	2
$\vdots$	$\vdots$	$\vdots$

Then we have that:

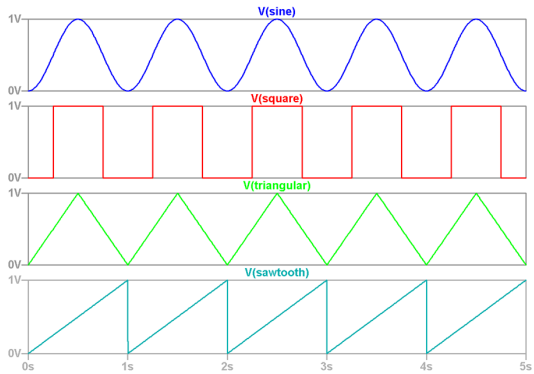
$$b_m = \begin{cases} \frac{4}{m\pi} & m \text{ odd} \\ 0 & m \text{ even} \end{cases}$$

Thus, the Fourier series for the square wave function is:

$$f(t) = \frac{4}{\pi} \left(\sin(t) + \frac{1}{3} \sin(3t) + \frac{1}{5} \sin(5t) + \cdots \right)$$

Continuity, differentiability, and rate of decay of terms

Consider different waves: a square wave, triangle wave, and a wave consisting of parabolas.



The square wave is discontinuous, the triangle wave is continuous but not differentiable, and the parabola wave is continuous and differentiable.

Integrating the formula for a coefficient by parts:

$$\begin{aligned} a_m &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(mt)dt \\ &= \frac{1}{\pi}\left[f(t)\frac{\sin(mt)}{m}\right]_{-\pi}^{\pi} - \frac{1}{\pi}\int_{-\pi}^{\pi} f'(t)\frac{\sin(mt)}{m}dt \end{aligned}$$

If f is continuous, then the first term is zero, and we have:

$$\begin{aligned} a_m &= -\frac{1}{m\pi}\int_{-\pi}^{\pi} f'(t)\sin(mt)dt \\ &= -\frac{1}{m}b'_m \end{aligned}$$

Then if $f(t)$ is twice differentiable, another integration is possible.

Notice that we are repeatedly integrating by parts, and each time we do so, we get an extra factor of $\frac{1}{m}$ in the denominator. Thus, if f is k times differentiable, then we have:

$$a_m = \frac{(-1)^k}{m^k}b_m^{(k)}$$

Note that in the $b_m^{(k)}$ term, there is a "hidden" $1/m$ term, since b_m is defined as an integral of $f(t)\sin(mt)$, and integrating by parts will introduce a factor of $1/m$ each time. Thus, we have:

$$a_m = \frac{(-1)^k}{m^{k+1}}b_m^{(k+1)}$$

This means that the Fourier coefficients decay at a rate of $\frac{1}{m^{k+1}}$ if the function is k times differentiable. In particular, if a function is infinitely differentiable, then its Fourier coefficients decay faster than any polynomial rate.

For example, the Fourier coefficients of the square wave function decay at a rate of $\frac{1}{m}$, since the square wave is not continuous. The Fourier coefficients of the triangle wave function decay at a rate of $\frac{1}{m^2}$, since the triangle wave is continuous but not differentiable. The Fourier coefficients of the parabola wave function decay at a rate of $\frac{1}{m^3}$, since the parabola wave is continuous and differentiable.

Note:-

Piecewise continuous functions are functions that are continuous on each piece of their domain, but may have a finite number of jump discontinuities. For example, the square wave function is piecewise continuous, as it is continuous on the intervals $(-\pi, 0)$ and $(0, \pi)$, but has a jump discontinuity at $t = 0$.
We can define the left and right limits of a function f at a point a as follows:

$$f(a^-) = \lim_{t\to a^-} f(t), \quad f(a^+) = \lim_{t\to a^+} f(t)$$

A **nice function** is defined if we have $f(a) = \frac{1}{2}\left(f(a^-) + f(a^+)\right)$ for all a. In other words, the function is

defined at the point of discontinuity as the average of the left and right limits. If a function is nice, then its Fourier series converges to the function at all points.

The Gibb's phenomenon is the overshoot that occurs near a jump discontinuity in the Fourier series approximation of a function. The overshoot does not go away as we include more terms in the Fourier series, and it approaches a fixed percentage of the jump discontinuity as the number of terms goes to infinity.

4.1.4 Manipulating Fourier series

$$\boxed{\sin(\theta + \frac{\pi}{2}) = \cos(\theta), \quad \cos(\theta + \frac{\pi}{2}) = -\sin(\theta)}$$

- We can take linear combinations of Fourier series.
- We can shift a Fourier series by a constant.
- We can stretch a Fourier series by a constant.

Example 4.1.1 (Linear combinations of Fourier series)

We can take linear combinations of Fourier series by taking linear combinations of the Fourier coefficients. For example, if we have two functions f and g with Fourier coefficients a_m^f, b_m^f and a_m^g, b_m^g respectively, then the Fourier coefficients of the function $h = af + bg$ are given by:

$$\begin{aligned} f(t) &= 1 + 2\mathrm{sq}(t) \\ &= 1 + \frac{8}{\pi}\left(\sin(t) + \frac{1}{3}\sin(3t) + \frac{1}{5}\sin(5t) + \cdots\right) \\ &= 1 + \frac{8}{\pi}\left(\sin\left(t + \frac{\pi}{2}\right) + \frac{1}{3}\sin\left(3t + \frac{\pi}{2}\right) + \frac{1}{5}\sin\left(5t + \frac{\pi}{2}\right) + \cdots\right) \\ &= 1 + \frac{8}{\pi}\left(\cos(t) + \frac{1}{3}\cos(3t) + \frac{1}{5}\cos(5t) + \cdots\right) \end{aligned}$$

Example 4.1.2 (Shifting a Fourier series)

We can shift a Fourier series by a constant by using the angle addition formulas for sine and cosine. For example, if we have a function $f(t)$ with Fourier coefficients a_m and b_m, then the Fourier coefficients of the function $g(t) = f(t - \frac{\pi}{2})$ are given by:

$$\begin{aligned} f(t) &= \mathrm{sq}(t - \frac{\pi}{2}) \\ &= \frac{4}{\pi}\left(\sin\left(t - \frac{\pi}{2}\right) + \frac{1}{3}\sin\left(3t - \frac{3\pi}{2}\right) + \cdots\right) \\ &= \frac{4}{\pi}\left(-\cos(t) + \frac{1}{3}\cos(3t) + \cdots\right) \end{aligned}$$

Example 4.1.3 (Stretching a Fourier series)

We can stretch a Fourier series by a constant by using the angle addition formulas for sine and cosine. For example, if we have a function $f(t)$ with Fourier coefficients a_m and b_m, then the Fourier coefficients of the function $g(t) = f(2t)$ are given by:

$$\begin{aligned} f(t) &= \mathrm{sq}(2t) \\ &= \frac{4}{\pi}\left(\sin(2t) + \frac{1}{3}\sin(6t) + \cdots\right) \end{aligned}$$

Then this function has period π, since the period of $\sin(2t)$ is π.

4.1.5 Differentiating and integrating Fourier series

We can differentiate a Fourier series term by term, since the Fourier series converges uniformly. For example, if we have a function $f(t)$ with Fourier coefficients a_m and b_m, then the Fourier coefficients of the function $g(t) = f'(t)$ are given by:

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos(mt) + b_m \sin(mt))$$
$$f'(t) = \sum_{m=1}^{\infty} (-m a_m \sin(mt) + m b_m \cos(mt))$$

Similarly, we can integrate a Fourier series term by term. For example, if we have a function $f(t)$ with Fourier coefficients a_m and b_m, then the Fourier coefficients of the function $g(t) = \int f(t)dt$ are given by:

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos(mt) + b_m \sin(mt))$$
$$\int f(t)dt = \frac{a_0}{2}t + \sum_{m=1}^{\infty} \left(\frac{a_m}{m} \sin(mt) - \frac{b_m}{m} \cos(mt) \right)$$

Example 4.1.4 (Differentiating the square wave function)
We can differentiate the square wave function term by term to get:

$$f(t) = \frac{4}{\pi} \left(\sin(t) + \frac{1}{3} \sin(3t) + \cdots \right)$$
$$f'(t) = \frac{4}{\pi} (\cos(t) + \cos(3t) + \cdots)$$

Then we have that $f'(t)$ is a sum of cosine functions, which is a periodic function with period 2π.

4.1.6 Parseval's theorem and Basel problem

We recall that the square wave function is given by:

$$f(t) = \frac{4}{\pi} \left(\sin(t) + \frac{1}{3} \sin(3t) + \cdots \right)$$

Then we have that:

$$f(t) = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \sin(nt)$$

Theorem 4.1.1 Parseval's theorem
If $f(t)$ is a periodic function with Fourier coefficients a_m and b_m, then we have:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt = \frac{a_0^2}{2} + \sum_{m=1}^{\infty} \left(a_m^2 + b_m^2 \right)$$

Proof.: We use the orthogonality of the sine and cosine functions to compute the integral. Anytime we have a $\cos(mt)\cos(nt)$ or $\sin(mt)\sin(nt)$ term, we get zero unless $m = n$. Cross-terms with $\cos(mt)\sin(nt)$ vanish. Any term with a_0 will vanish when multiplied by $\cos(mt)$ or $\sin(mt)$, since the average value of $\cos(mt)$ and $\sin(mt)$ is zero.

Thus, we have:

$$\begin{aligned}
\frac{1}{\pi}\int_{-\pi}^{\pi}|f(t)|^2dt &= \frac{1}{\pi}\int_{-\pi}^{\pi}\left(\frac{a_0}{2}+\sum_{m=1}^{\infty}\left(a_m\cos(mt)+b_m\sin(mt)\right)\right)^2dt \\
&= \frac{1}{\pi}\int_{-\pi}^{\pi}\left(\frac{a_0^2}{4}+\sum_{m=1}^{\infty}\left(a_m^2\cos^2(mt)+b_m^2\sin^2(mt)\right)+\text{cross-terms}\right)dt \\
&= \frac{a_0^2}{2}+\sum_{m=1}^{\infty}\left(a_m^2+b_m^2\right)
\end{aligned}$$

☺

Example 4.1.5 (Basel problem)

We can use Parseval's theorem to solve the Basel problem, which asks for the value of the series $\sum_{n=1}^{\infty}\frac{1}{n^2}$. First, let's consider the square wave function again:

$$f(t)=\frac{4}{\pi}\sum_{n\text{ odd}}\frac{1}{n}\sin(nt)$$

From Parseval's theorem, we have:

$$\begin{aligned}
\frac{1}{\pi}\int_{-\pi}^{\pi}|f(t)|^2dt &= \sum_{n\text{ odd}}\left(\frac{4}{\pi}\frac{1}{n}\right)^2 \\
&= \frac{16}{\pi^2}\sum_{n\text{ odd}}\frac{1}{n^2}
\end{aligned}$$

On the other hand, if we take $\int_{-\pi}^{\pi}|f(t)|^2dt$, we can compute this integral directly. Since $f(t)$ is a square wave, we have:

$$\begin{aligned}
\frac{1}{\pi}\int_{-\pi}^{\pi}|f(t)|^2dt &= \frac{1}{\pi}\left(\int_{-\pi}^{0}1^2dt+\int_{0}^{\pi}1^2dt\right) \\
&= \frac{1}{\pi}(0-(-\pi)+\pi-0) \\
&= 2
\end{aligned}$$

Thus, we have:

$$\begin{aligned}
2 &= \frac{16}{\pi^2}\sum_{n\text{ odd}}\frac{1}{n^2} \\
\sum_{n\text{ odd}}\frac{1}{n^2} &= \frac{\pi^2}{8}
\end{aligned}$$

Now we can express the sum over odd integers in terms of the sum over all integers:

$$\begin{aligned}
\sum_{n=1}^{\infty}\frac{1}{n^2} &= \sum_{n\text{ odd}}\frac{1}{n^2}+\sum_{n\text{ even}}\frac{1}{n^2} \\
&= \sum_{n\text{ odd}}\frac{1}{n^2}+\frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} \quad \text{since } \frac{1}{2^2}+\frac{1}{4^2}+\cdots=\frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} \\
&= \frac{\pi^2}{8}+\frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2}
\end{aligned}$$

Solving for $\sum_{n=1}^{\infty}\frac{1}{n^2}$, we get:

$$\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$$

Example 4.1.6 (Basel problem with the triangle wave)

Suppose that we want to compute $\sum_{n=1}^{\infty} \frac{1}{n^4}$. From Parseval's theorem, we know that we could get the odd part $\sum_{n \text{ odd}} \frac{1}{n^4}$ if we had some function with Fourier coefficients that decay at a rate of $\frac{1}{n^2}$.

We noticed that integrating a Fourier series term by term will introduce a factor of $\frac{1}{n}$ in the denominator, so if we start with a function whose Fourier coefficients decay at a rate of $\frac{1}{n}$, then integrating it will give us a function whose Fourier coefficients decay at a rate of $\frac{1}{n^2}$. The square wave function has Fourier coefficients that decay at a rate of $\frac{1}{n}$, so we can integrate the square wave function to get a function whose Fourier coefficients decay at a rate of $\frac{1}{n^2}$. Integrating the square wave function, we get:

$$\begin{aligned} f(t) &= \int \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \sin(nt) dt \\ &= \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n^2} (-\cos(nt)) + C \end{aligned}$$

And if we were to integrate the square wave geometrically in a piecewise way, we'd find that the resulting function is a triangle wave, defined as:

$$f(x) = \begin{cases} -x & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

Then we have that the Fourier coefficients of the triangle wave function are given by:

$$\begin{aligned} a_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(mt) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} |t| \cos(mt) dt \quad \text{even function} \\ &= \frac{2}{\pi} \int_{0}^{\pi} t \cos(mt) dt \\ &= \frac{2}{\pi} \left[t \frac{\sin(mt)}{m} \right]_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} \frac{\sin(mt)}{m} dt \\ &= \frac{2}{\pi} \left[\frac{\pi \sin(m\pi)}{m} - 0 \right] + \frac{2}{m^2 \pi} \left[-\cos(mt) \right]_0^{\pi} \\ &= \frac{2}{m^2 \pi} \left[1 - \cos(m\pi) \right] \\ &= \begin{cases} \frac{4}{m^2 \pi} & m \text{ odd} \\ 0 & m \text{ even} \end{cases} \end{aligned}$$

Applying Parseval's theorem, we have:

$$\begin{aligned} \frac{1}{\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt &= \frac{1}{2} a_0^2 + \sum_{n \text{ odd}} \left(\frac{4}{n^2 \pi} \right)^2 \\ &= \frac{\pi^2}{2} + \frac{16}{\pi^2} \sum_{n \text{ odd}} \frac{1}{n^4} \end{aligned}$$

With the LHS being

$$\begin{aligned} \frac{1}{\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt &= \frac{1}{\pi} \left(\int_{-\pi}^{0} t^2 dt + \int_0^{\pi} t^2 dt \right) \\ &= \frac{1}{\pi} \left(\frac{\pi^3}{3} + \frac{\pi^3}{3} \right) \\ &= \frac{2\pi^2}{3} \end{aligned}$$

Solving for $\sum_{n \text{ odd}} \frac{1}{n^4}$, we get:

$$\sum_{n \text{ odd}} \frac{1}{n^4} = \frac{\pi^4}{96}$$

Then we can express the sum over odd integers in terms of the sum over all integers:

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{1}{n^4} &= \sum_{n \text{ odd}} \frac{1}{n^4} + \sum_{n \text{ even}} \frac{1}{n^4} \\
&= \sum_{n \text{ odd}} \frac{1}{n^4} + \frac{1}{16} \sum_{n=1}^{\infty} \frac{1}{n^4} \quad \text{since } \frac{1}{2^4} + \frac{1}{4^4} + \cdots = \frac{1}{16} \sum_{n=1}^{\infty} \frac{1}{n^4} \\
&= \frac{\pi^4}{96} + \frac{1}{16} \sum_{n=1}^{\infty} \frac{1}{n^4}
\end{aligned}$$

Solving for $\sum_{n=1}^{\infty} \frac{1}{n^4}$, we get:

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

4.1.7 Applying Fourier series to ODEs

Definition 4.1.1: Superposition principle

If f and g are solutions to a linear homogeneous ordinary differential equation, then any linear combination of f and g is also a solution to the same differential equation.

This principle allows us to solve each term in the Fourier series separately, and then add up the solutions to get the solution to the original problem.

Example 4.1.7 (Square wave forcing function)

Consider some ODE of the form:

$$\ddot{x} + \omega_m^2 x = \text{sq}(t)$$

We can express the square wave function as a Fourier series:

$$\text{sq}(t) = \frac{4}{\pi} \left(\sin(t) + \frac{1}{3} \sin(3t) + \cdots \right)$$

Recall that to find $x_p(t)$, we could apply the ERF to each term (assuming no resonance). Then we obtain:

$$x_p(t) = \left(\frac{4}{\pi} \right) \left[\frac{1}{\omega_m^2 - 1} \sin(t) + \frac{1}{3} \frac{1}{\omega_m^2 - 9} \sin(3t) + \cdots \right]$$

What if we are close to resonance, that is, there is one term where $\omega_m^2 - n^2$ is close to zero? That term would dominate the solution.

Let r be some forcing frequency, which can also be complex. We say that $p(r) = \epsilon$ which is very small, so r is close to a root of p. Then we have that the particular solution to the ODE is given by:

$$x_p(t) = \frac{1}{\epsilon} \sin(rt)$$

Example 4.1.8 (Particular solution as a sum of cosines)

We solve:

$$\ddot{x}_n + 2\dot{x}_n + 9x_n = \cos(nt)$$

For $n = 0$, we have $x_0 = \frac{1}{9}$.
For $n > 0$, we can complexify:

$$\ddot{z}_n + 2\dot{z}_n + 9z_n = e^{int}$$

Inputting into the ERF:

$$z_n(t) = \frac{1}{9 - n^2 + 2in} e^{int}$$

Let R_n be the magnitude of the complex number given by $p(in)^{-1}$, and let ϕ_n be the argument.
Then we have:

$$R_n^{-1} = \sqrt{(9 - n^2)^2 + 4n^2}$$

$$\phi_n = \tan^{-1}\left(\frac{2n}{9 - n^2}\right)$$

Then we can write $z_n(t)$ as:

$$z_n(t) = R_n e^{i(nt - \phi_n)}$$

Then we can take the real part to get $x_n(t)$:

$$x_n(t) = R_n \cos(nt - \phi_n)$$

This enables us to easily take the real part when solving for the particular solution for a Fourier series consisting of a sum of cosines.

$$x_p(t) = R_0 \cos(-\phi_0) + \sum_{n=1}^{\infty} R_n \cos(nt - \phi_n)$$

4.1.8 Complex replacement

Without loss of generality and over the period $[-L, L]$, we can also write a Fourier series in terms of exponentials.

Recall that we have a Fourier series over $[-L, L]$ written as:

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \left(a_m \cos\left(\frac{m\pi t}{L}\right) + b_m \sin\left(\frac{m\pi t}{L}\right)\right)$$

Using this as a forcing function:

$$p(D)x = f(t)$$

Using the complex replacement, we can write this as:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} c_n e^{in\pi t/L}$$

Then we can write $z_p(t)$ as the sum of two particular solutions:

$$z_p(t) = z_0(t) + \sum_{n=1}^{\infty} z_n(t)$$

First, we can solve for $z_0(t)$:

$$p(D)z_0(t) = \frac{a_0}{2}$$
$$z_0(t) = \frac{a_0}{2p(0)}$$

If $p(0) = 0$, this implies that the coefficient before the x term in $p(D)$ is zero. Apply reduction of order.

Next, we can solve for $z_n(t)$. But we also need to check for resonance.

Resonance occurs when $p(\frac{in\pi}{L}) = 0$. This occurs when $\frac{in\pi}{L}$ is a root of p. We call this n_R, or the resonant frequency.

In general, we differentiate multiple times:

$$p(\frac{i\pi n}{L}) = p'(\frac{i\pi n}{L}) = \cdots = p^{(k-1)}(\frac{i\pi n}{L}) = 0$$

Until we find some k such that $p^{(k)}(\frac{i\pi n}{L}) \neq 0$.

For:

$$\boxed{z_{p,n\neq n_R} = \frac{c_n}{p(\frac{in\pi}{L})} e^{\frac{in\pi}{L}t}}$$

For the resonant term, we have:

$$\boxed{z_{p,n=n_R} = \frac{c_n}{p^{(k)}(\frac{in\pi}{L})} t^k e^{\frac{in_R\pi}{L}t}}$$

Example 4.1.9

Consider the ODE:

$$\ddot{x} + 9x = t_r(t)$$

where $t_r(t)$ is the triangle wave function defined as:

$$t_r(t) = \begin{cases} -t & -\pi < t < 0 \\ t & 0 < t < \pi \end{cases}$$

Representing as a Fourier series, we have:

$$t_r(t) = \frac{\pi}{2} - \frac{8}{\pi}\left(\frac{1}{1^2}\cos(t) + \frac{1}{3^2}\cos(3t) + \cdots\right)$$

This has characteristic polynomial:

$$p(r) = r^2 + 9$$

We find that the roots of the homogeneous equation are $\pm 3i$.

Now, we complexify to obtain:

$$\ddot{z} + 9z = \frac{\pi}{2} - \frac{8}{\pi}\sum_{n \text{ odd}} \frac{1}{n^2} e^{int}$$

Resonance occurs at the $n = 3$ term. Check $p(3i) = 0$, then $p'(3i) = 6i \neq 0$. Hence:

$$\begin{aligned} z_{p,m}(t) &= \frac{1}{2\cdot 9} - \frac{4}{\pi^2}\left(\frac{e^{it}}{p(i)} + \frac{1}{9}\cdot\frac{1}{p'(3i)} te^{3it} + \frac{1}{25}\cdot\frac{1}{p(5i)} e^{5it} + \cdots\right) \\ &= \frac{1}{18} - \frac{4}{\pi^2}\left(\frac{e^{it}}{8} + \frac{1}{54} te^{3it} - \frac{1}{400} e^{5it} + \cdots\right) \end{aligned}$$

All terms will have real part cos except the resonant one:

$$\begin{aligned} \mathrm{Re}\left(\frac{1}{54i} te^{3it}\right) &= \mathrm{Re}\left(\frac{-i}{54} e^{3it}\right) \\ &= \frac{1}{54}\sin(3t) \end{aligned}$$

Then $x_p(t)$:

$$\boxed{x_p(t) = \frac{1}{18} - \frac{4}{\pi^2}\left(\frac{1}{8}\cos(t) + \frac{1}{54} t\sin(3t) - \frac{1}{400}\cos(5t) + \cdots\right)}$$

Chapter 5

Laplace transforms

5.1 Heaviside function

So far, we have dealt with so-called *nice functions*. Consider the step function:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

where $u(0)$ is undefined. This function is called the **Heaviside function**.

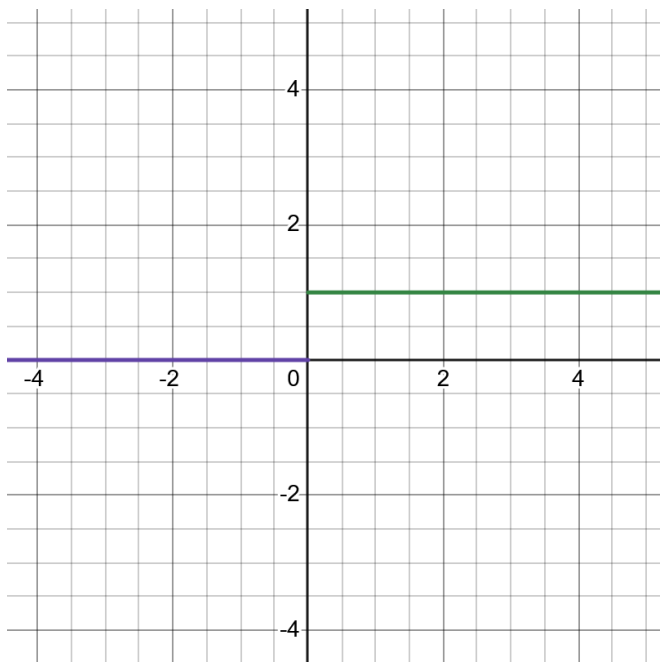


Figure 5.1: Graph of the Heaviside function

We can apply a shift to the Heaviside function such that it "turns on" at $t = a$:

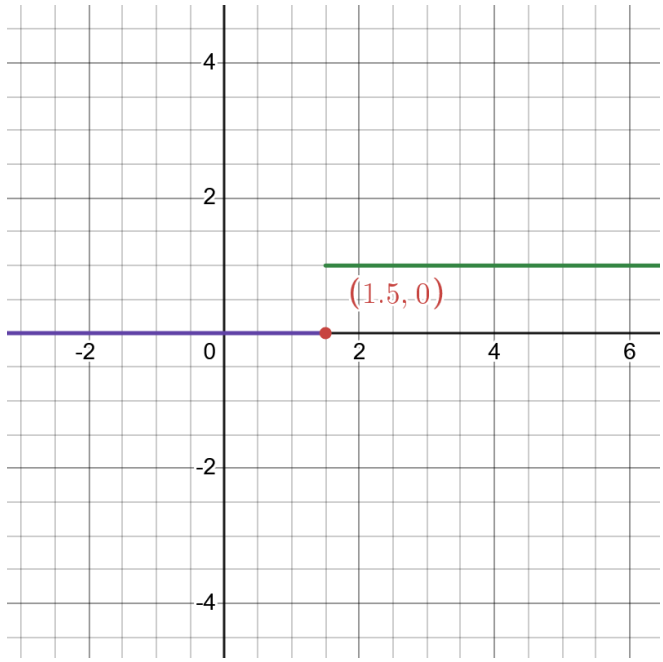


Figure 5.2: Graph of the shifted Heaviside function

Finally, if we take two shifted Heaviside functions such that $a < b$, we get a function that resembles the following:

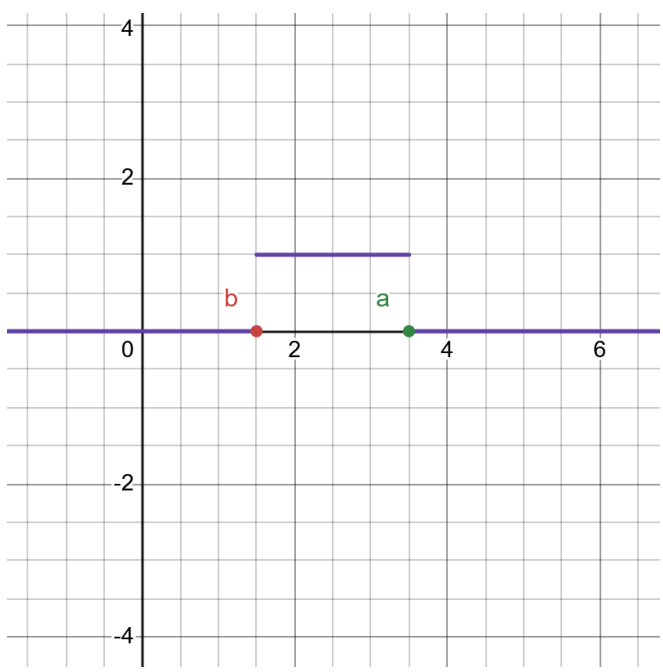


Figure 5.3: Graph of the shifted Heaviside function

We can see that this becomes useful to restrict the range of a function $f(t)$ to a certain interval. If the function is only "on" with $t \in (a, b)$, then we can write $f(t) = f(t)(u(t-a) - u(t-b))$.

5.1.1 Bank account example

Let's assume we have a bank account for which the change in our balance is modeled by the following:

$$x(t + \Delta t) = x(t) + Ix(t)\Delta t$$

where $x(t)$ is the amount current in the account, Δt is some time over which interest is accumulated, and I is the interest rate.

Remark: This discrete model is extremely similar to Euler's forward method where $f(x+h) = x + h \cdot f(x)$.

At $t = 0$, we have that $\dot{x} = Ix$. That is, the change in our balance is proportional to the amount currently in the account. Let us define some rate $q(t)$ at which money is deposited into the account. Note that this is distinct from the interest rate I which is constant. Then, we have that $\dot{x} - Ix = q(t)$. We must now attempt to determine the cumulative total amount of money $Q(t)$ that has been deposite. It is clear that:

$$\dot{Q} = q(t) \Longrightarrow Q(t) = \int_0^t q(\bar{t})d\bar{t}$$

Now consider the case where a single \$2000 occurs at $t = 1$, while we are also depositing money at a rate of \$1000 per year.

Intuitively, we would write $Q(t)$ as:

$$Q(t) = 1000t + 2000u(t-1)$$

We note that the Heaviside function is perfect to represent this situation: we "switch on" the deposit at $t = 1$ and the money remains in the account.

However, we are presented with the challenge of differentiating this function to derive $q(t)$. No such function exists for $u(t)$, but there is a "function" (which is actually a distribution) that satisifes the following properties that we'd expect for such a derivative:

- We know that the integral of this derivative must be equal to 1 to be equivalent to $u(t)$.
- The derivative must be zero for $t < 0$ since $u(t)$ is constant for $t < 0$, and the derivative must also be zero for $t > 0$ since $u(t)$ is constant for $t > 0$ - the only point at which $u(t)$ is not constant is at $t = 0$.

This "function" is called the **Dirac delta function** $\delta(t)$, and it is defined as follows:

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

In the example above, we have that $q(t) = 1000 + 2000\delta(t-1)$.

We could even consider a case where at $t = 1$, we deposit \$1000, and at $t = 2$, we withdraw \$500. In this case, we have that $q(t) = 1000\delta(t-1) - 500\delta(t-2)$.

In particular, we visualize the Dirac delta function like this:

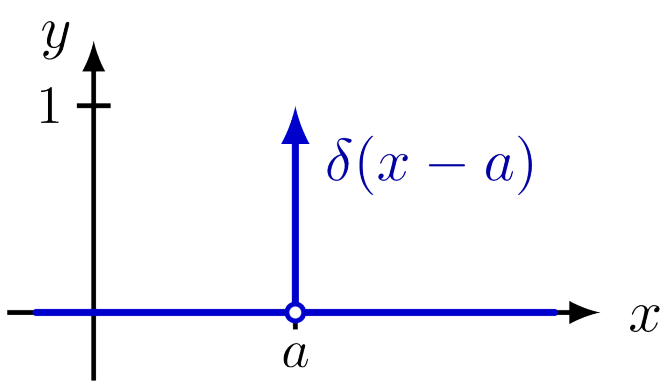


Figure 5.4: Graph of the Dirac delta function

As convention, we write the a beside the delta function to indicate the integral of the delta function:

$$\int_{-\infty}^{\infty} a\delta(t)dt = a$$

Now recall piecewise smooth functions. These functions are broken up into smooth pieces that have limitson the right and on th eleft that exist at each break. These are therefore "nice" functions that switch on or off with different applications of $u(t)$.

For example, consider the function:

$$f(t) = \begin{cases} t, & t < 1 \\ 1 - \sqrt{t}, & 1 < t < 2 \\ t - 2, & t > 2 \end{cases}$$

This function appears like this:

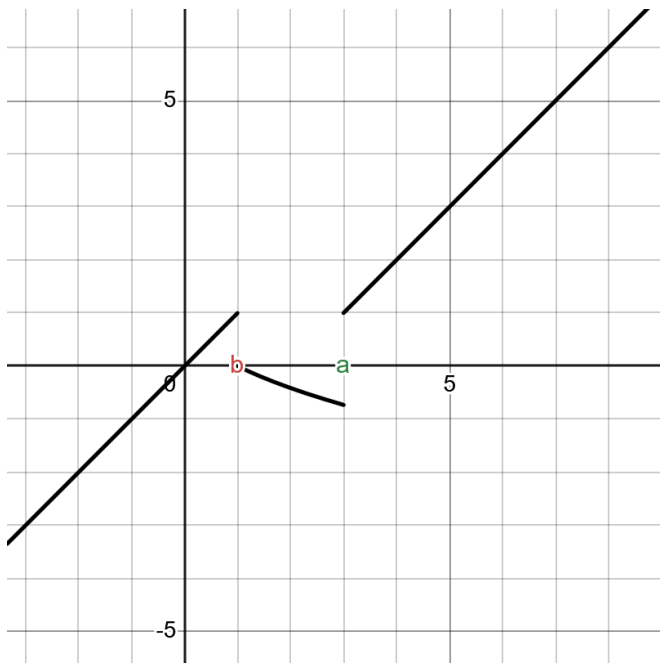


Figure 5.5: Graph of a piecewise smooth function

We can write this function in terms of the Heaviside function as follows:

$$f(t) = t + (1 - \sqrt{t} - t)u(t-1) + (t - 2 - (1 - \sqrt{t}))u(t-2)$$

In general, we can "turn on" a function $f(t)$ at $t = a$ and "turn off" the function at $t = b$ by writing $f(t)(u(t-a) - u(t-b))$. We can also "turn on" a function at $t = a$ and leave it on for all $t > a$ by writing $f(t)u(t-a)$. Then for a piecewise smooth function, we can write it as a sum of functions that are "turned on" at different points. For example, if we have a piecewise smooth function with break points at $t = a_1, a_2, \ldots, a_n$, then we can write the function as:

$$f(t) = f_1(t) + (f_2(t) - f_1(t))u(t-a_1) + (f_3(t) - f_2(t))u(t-a_2) + \ldots + (f_n(t) - f_{n-1}(t))u(t-a_{n-1})$$

Basically, at a_1, we "turn on" the function $f_2(t)$ with multiplication by $u(t-a_1)$ while the subtraction of $f_1(t)u(t-a_1)$ at the same time "turns off" the function $f_1(t)$, and this process continues for each break point.

Definition 5.1.1: Generalized derivative

The **generalized derivative** of a function $f(t)$ is defined as the limit of the difference quotient as h approaches zero, where the difference quotient is given by:

$$\frac{f(t+h)-f(t)}{h}$$

This definition allows us to extend the concept of differentiation to functions that may not be differentiable in the traditional sense, such as piecewise smooth functions or functions with discontinuities. The generalized derivative can be used to analyze the behavior of such functions and to solve differential equations that involve them.

In particular, given a piecewise smooth function $Q(t)$, we can now find a derivative at point a by:

$$\dot{Q}(t) = (Q(a^+) - Q(a^-))\delta(t-a)$$

where $Q(a^+)$ and $Q(a^-)$ are the right and left limits of $Q(t)$ at $t = a$, respectively.

We denote this as:

$$\boxed{[Q]_a \delta(t-a)}$$

Example 5.1.1 (Example)

Consider the function:

$$Q(t) = \begin{cases} 0, & t < 1 \\ 1000t, & 1 < t < 2 \\ 2000, & t > 2 \end{cases}$$

Find $\dot{Q}(t)$.

Solution: We can write $Q(t)$ in terms of the Heaviside function as follows:

$$Q(t) = 1000tu(t-1) + (2000 - 1000t)u(t-2) = 1000tu(t-1) + 2000u(t-2) - 1000tu(t-2)$$

We have three terms to differentiate.

$$\begin{aligned} \frac{d}{dt}1000tu(t-1) &= 1000u(t-1) + 1000t\delta(t-1) \\ \frac{d}{dt}2000u(t-2) &= 2000\delta(t-2) \\ \frac{d}{dt}1000tu(t-2) &= 1000u(t-2) + 1000t\delta(t-2) \end{aligned}$$

Now we can combine these three terms to get:

$$\begin{aligned} \dot{Q}(t) &= 1000u(t-1) + 1000t\delta(t-1) + 2000\delta(t-2) - 1000u(t-2) - 1000t\delta(t-2) \\ &= 1000u(t-1) - 1000u(t-2) + 1000t\delta(t-1) + (2000 - 1000t)\delta(t-2) \end{aligned}$$

Now we use the property that $f(t) = f(a)\delta(t-a)$ to simplify the delta function terms:

$$\begin{aligned} \dot{Q}(t) &= 1000u(t-1) - 1000u(t-2) + 1000t\delta(t-1) + (2000 - 1000t)\delta(t-2) \\ &= 1000u(t-1) - 1000u(t-2) + 1000 \cdot 1 \cdot \delta(t-1) + (2000 - 1000 \cdot 2)\delta(t-2) \\ &= 1000u(t-1) - 1000u(t-2) + 1000\delta(t-1) + 0 \cdot \delta(t-2) \\ &= 1000u(t-1) - 1000u(t-2) + 1000\delta(t-1) \end{aligned}$$

This can be interpeted as follows: at $t = 1$, we have a deposit of \$1000, which is represented by the delta function term. For $t > 1$, we have a continuous deposit of \$1000 per year, which is represented by the

first Heaviside function term. Finally, at $t = 2$, we stop depositing money, which is represented by the second Heaviside function term.

Finally, we note that we can shift the delta function as follows:

$$\delta(t-a) = \frac{d}{dt}u(t-a)$$

And recalling that:

$$\int_{-\infty}^{\infty} \delta(t-1)dt = \int_{-\infty}^{\infty} \delta(t')dt' = 1$$

Applying this to the integral of the delta function multiplied by some function $f(t)$, we have:

$$\int_{-\infty}^{\infty} f(t')\delta(t'-t)dt' = f(t)\int_{-\infty}^{\infty} \delta(t'-t)dt'$$

We can do this because the delta function is zero everywhere except at $t' = t$, so we can factor out $f(t)$ from the integral. It effectively has a single value, and hence can be treated as a constant. Now, since the integral of the delta function is 1, we have:

$$\begin{aligned}\int_{-\infty}^{\infty} f(t')\delta(t'-t)dt' &= f(t)\cdot 1 \\ &= f(t)\end{aligned}$$

5.2 Unit response function

5.2.1 First-order linear systems

Let there be a system described by:

$$\dot{x} + kx = r \cdot u(t)$$

That is, before $t = 0$, the system is at rest, and at $t = 0$, we "turn on" the system with a constant input of r. We want to find the response of the system to this input.

This can easily be solved using the integrating factor method with $\mu(t) = e^{kt}$, which gives us:

$$\begin{aligned}e^{kt}\dot{x} + ke^{kt}x &= re^{kt}u(t) \\ \frac{d}{dt}(e^{kt}x) &= re^{kt}u(t) \\ e^{kt}x &= r\int_0^t e^{k\bar{t}}d\bar{t} \\ x &= re^{-kt}\int_0^t e^{k\bar{t}}d\bar{t} \\ &= re^{-kt}\cdot\frac{e^{kt}-1}{k} \\ &= \frac{r}{k}(1-e^{-kt}) \quad \text{for } t > 0\end{aligned}$$

Definition 5.2.1: Unit impulse response function

The **unit impulse response function** of a system is the response of the system to a unit impulse input ($\delta(t)$).

For a first-order linear system described by the equation $\dot{x} + kx = r \cdot u(t)$, the unit response function is given by:

$$\boxed{v(t) = \frac{r}{k}(1-e^{-kt})u(t)}$$

We now plug $v(t)$ into the original equation (letting $r = 1$):

$$\dot{v} + kv = u(t) \quad \text{w/} \quad v(0^-) = 0$$

We can now solve for $\dot{v}$:

$$\begin{aligned}
\dot{v} &= u(t) - kv \\
&= u(t) - k \cdot \frac{1}{k}(1 - e^{-kt})u(t) \\
&= u(t) - (1 - e^{-kt})u(t) \\
&= e^{-kt}u(t)
\end{aligned}$$

Note that at $t = 0^+$, we have:

$$\begin{aligned}
\dot{v}(0^+) &= e^{-k \cdot 0}u(0) \\
&= 1 \cdot 1 \\
&= 1
\end{aligned}$$

And we know that $u(0^+) = 1$, so basically we have that:

$$\boxed{\dot{v}(0^+) = u(0^+)}$$

Example 5.2.1 (Basic unit impulse response function)
Consider the system described by the following equation:

$$\dot{x} + kx = \delta(t)$$

Find the response of the system to this input.

Solution: Solving the homogeneous equation $\dot{x} + kx = 0$ gives us that $x_h(t) = Ce^{-kt}$. Now we can use the method of variation of parameters to find a particular solution. Let $x_p(t) = C(t)e^{-kt}$, then we have:

$$\begin{aligned}
\dot{x}_p + kx_p &= \dot{C}e^{-kt} - kCe^{-kt} + kCe^{-kt} \\
&= \dot{C}e^{-kt} \\
&= \delta(t)
\end{aligned}$$

Now we can solve for $\dot{C}$:

$$\begin{aligned}
\dot{C} &= e^{kt}\delta(t) \\
&= e^{k \cdot 0}\delta(t) \\
&= 1 \cdot \delta(t) \\
&= \delta(t)
\end{aligned}$$

Now we can integrate both sides to find $C(t)$:

$$\begin{aligned}
C(t) &= \int_{-\infty}^{t} \delta(\bar{t})d\bar{t} \\
&= u(t)
\end{aligned}$$

Finally, we can write the particular solution as follows:

$$\begin{aligned}
x_p(t) &= C(t)e^{-kt} \\
&= u(t)e^{-kt}
\end{aligned}$$

Thus, the general solution to the nonhomogeneous equation is given by:

$$\begin{aligned} x(t) &= x_h(t) + x_p(t) \\ &= Ce^{-kt} + u(t)e^{-kt} \end{aligned}$$

Now we can apply the initial condition $x(0^-) = 0$ to solve for C:

$$\begin{aligned} x(0^-) &= Ce^{-k\cdot 0} + u(0^-)e^{-k\cdot 0} \\ &= C \cdot 1 + 0 \cdot 1 \\ &= C \end{aligned}$$

Thus, we have that $C = 0$, and the response of the system to the input $\delta(t)$ is given by:

$$\boxed{x(t) = u(t)e^{-kt}}$$

We can further characterize the unit impulse function. For example, since the function is only "on" from $t > 0$, we have that $w(0^-) = 0$ and $w(0^+) = 1$.

Then, because $\delta(0^+) = 0$, we can also find that:

$$\begin{aligned} \dot{w}(0^+) &= \frac{d}{dt}w(t)\bigg|_{t=0^+} \\ &= \frac{d}{dt}(u(t)e^{-kt})\bigg|_{t=0^+} \\ &= -ke^{-kt}u(t) + e^{-kt}\delta(t)\bigg|_{t=0^+} \\ &= -k \cdot 1 \cdot 1 + 1 \cdot 0 \\ &= -k \end{aligned}$$

5.2.2 Second-order linear systems

Consider the second-order linear system described by:

$$m\ddot{x} + kx = \delta(t)$$

That is, there is no damping in the system, and an increase in momentum occurs at $t = 0$ with no change in position.

We will use the **shrinking interval method** to solve this problem. We will first solve the homogeneous equation $m\ddot{x} + kx = 0$ to find the general solution, and then we will apply the initial conditions to find the particular solution.

Solving the homogeneous equation:

$$\begin{aligned} mr^2 + k &= 0 \\ r^2 &= -\frac{k}{m} \\ r &= \pm i\sqrt{\frac{k}{m}} \quad (\text{let } \omega_m = \sqrt{\frac{k}{m}}) \\ r &= \pm i\omega_m \end{aligned}$$

Then writing $x_h(t)$:

$$x_h(t) = c_1 \cos(\omega_m t) + c_2 \sin(\omega_m t)$$

To find the initial conditions, consider the behaviour of the system as t approaches zero from the left and the right.

As t approaches zero from the left, we have that $x(0^-) = 0$ and $\dot{x}(0^-) = 0$ since the system is at rest before $t = 0$.

We can integrate the equation $m\ddot{x} + kx = \delta(t)$ from $t = -\epsilon$ to $t = \epsilon$ for some small $\epsilon > 0$:

$$\int_{-\epsilon}^{\epsilon} m\ddot{x} + kx dt = \int_{-\epsilon}^{\epsilon} \delta(t) dt$$
$$m\dot{x}(\epsilon) - m\dot{x}(-\epsilon) + k\int_{-\epsilon}^{\epsilon} x dt = 1$$

Now, as ϵ approaches zero, we have that $\dot{x}(-\epsilon) \to 0$ and $\int_{-\epsilon}^{\epsilon} x dt \to 0$, so we can simplify the equation to get:

$$m\dot{x}(0^+) = 1$$
$$\dot{x}(0^+) = \frac{1}{m}$$

Now we can apply the initial conditions to the general solution to find the particular solution:

$$\begin{aligned} x(0^+) &= c_1 \cos(0) + c_2 \sin(0) \\ &= c_1 \end{aligned}$$

$$\begin{aligned} \dot{x}(0^+) &= -c_1\omega_m \sin(0) + c_2\omega_m \cos(0) \\ &= c_2\omega_m \\ &= \frac{1}{m} \\ c_2 &= \frac{1}{m\omega_m} \end{aligned}$$

Thus, the response of the system to the input $\delta(t)$ is given by:

$$\boxed{x(t) = \frac{1}{m\omega_m}\sin(\omega_m t)u(t)}$$

5.2.3 Generalization to n-th order linear systems

Consider the n-th order linear system described by:

$$a_n x^{(n)} + a_{n-1}x^{(n-1)} + \ldots + a_1\dot{x} + a_0 x = u(t)$$

We know that for $t < 0$, we have that $x(0^-) = 0$, $\dot{x}(0^-) = 0$, …, $x^{(n-1)}(0^-) = 0$ since the system is at rest before $t = 0$.

Applying the shrinking interval method, we can integrate the equation from $t = -\epsilon$ to $t = \epsilon$ for some small $\epsilon > 0$:

$$\int_{-\epsilon}^{\epsilon} a_n x^{(n)} + a_{n-1}x^{(n-1)} + \ldots + a_1\dot{x} + a_0 x dt = \int_{-\epsilon}^{\epsilon} u(t) dt$$
$$a_n x^{(n-1)}(\epsilon) - a_n x^{(n-1)}(-\epsilon) + a_{n-1}x^{(n-2)}(\epsilon) - a_{n-1}x^{(n-2)}(-\epsilon) + \ldots + a_1 x(\epsilon) - a_1 x(-\epsilon) + a_0\int_{-\epsilon}^{\epsilon} x dt = 1$$

Then all terms with $x^{(n-1)}(-\epsilon)$, $x^{(n-2)}(-\epsilon)$, …, $x(-\epsilon)$, and $\int_{-\epsilon}^{\epsilon} x dt$ will approach zero as ϵ approaches zero, so we can simplify the equation to get:

$$a_n x^{(n-1)}(0^+) + a_{n-1}x^{(n-2)}(0^+) + \ldots + a_1 x(0^+) = 1$$
$$a_n x^{(n-1)}(0^+) = 1$$
$$x^{(n-1)}(0^+) = \frac{1}{a_n}$$

So we have the following initial conditions:

- $x(0^+) = 0$
- $\dot{x}(0^+) = 0$
- $\dots$
- $x^{(n-2)}(0^+) = 0$
- $x^{(n-1)}(0^+) = \frac{1}{a_n}$

5.2.4 Unit step response function

Definition 5.2.2: Unit step response function

The **unit step response function** of a system is the response of the system to a unit step input ($u(t)$).

Example 5.2.2 (Finding the unit step response function)
Find the unit step response function of the system described by the following equation:

$$m\ddot{x} + kx = u(t)$$

Solution: We know that the homogeneous solution is:

$$x_h(t) = c_1 \cos(\omega_m t) + c_2 \sin(\omega_m t)$$

To find the particular solution where $t > 0$ implies $u(t) = 1$, we know that $x(0^+) = 0$, but we need the behaviour at $x(0^+)$ to find the particular solution. We can integrate the equation $m\ddot{x} + kx = u(t)$ from $t = -\epsilon$ to $t = \epsilon$ for some small $\epsilon > 0$:

$$\int_{-\epsilon}^{\epsilon} m\ddot{x} + kx dt = \int_{-\epsilon}^{\epsilon} u(t) dt$$
$$m\dot{x}(\epsilon) - m\dot{x}(-\epsilon) + k\int_{-\epsilon}^{\epsilon} x dt = 0$$

Now, as ϵ approaches zero, we have that $\dot{x}(-\epsilon) \to 0$ and $\int_{-\epsilon}^{\epsilon} x dt \to 0$, so we can simplify the equation to get:

$$m\dot{x}(0^+) = 0$$
$$\dot{x}(0^+) = 0$$

Then computing coefficients, we find that:

$$c_1 = \frac{-1}{k}$$
$$c_2 = 0$$

Then we get:

$$x(t) = \frac{1}{k}\cos(\omega_m t) - \frac{1}{k} \quad \text{for } t > 0$$

This function is only non-zero when $t > 0$, so we can multiply the function by $u(t)$ to get the final solution. Hence the **unit step response function** of the system is given by:

$$\boxed{x(t) = \frac{1}{k}(1 - \cos(\omega_m t))u(t)}$$

Recall that we can write the system as:

$$p(D)x = q(t)$$

where $p(D) = a_n D^n + a_{n-1} D^{n-1} + \ldots + a_1 D + a_0$ and $q(t)$ is some input function.

Then, differentiating both sides once:

$$p(D)\dot{x} = \dot{q}(t)$$

Now for the unit step function system:

$$p(D)x = u(t)$$

We can differentiate both sides once to get:

$$p(D)\dot{x} = \delta(t)$$

Thus, we can see that the unit step response function is the same as the unit impulse response function, except that the initial conditions are shifted by one order. For example, for a first-order linear system, the unit step response function will have an initial condition of $\dot{x}(0^+) = 0$ instead of $\dot{x}(0^+) = \frac{1}{a_1}$.

Also, given the unit step response function $v(t)$, we can find the unit impulse response function $w(t)$ by differentiating $v(t)$:

$$\boxed{\dot{v}(t) = w(t)}$$

5.3 Convolution

Let there be some system:

$$p(D)x = q(t)$$

where $q(t)$ is some input function, which is made of a sum of impulses. We can write $q(t)$ as a sum of impulses as follows:

$$q(t) = \int_{-\infty}^{\infty} q(\tau)\delta(t-\tau)d\tau$$

Recall the unit impulse response function $w(t)$. We can also write using linearity and δ function properties that:

$$p(D)aw(t-\tau) = a\delta(t-\tau)$$

Using the linearity of the system (and the fact that we can shift the delta function), we can write the response of the system to the input $q(t)$ as follows:

$$\begin{aligned} x(t) &= \sum_{-\infty}^{\infty} q(\tau)w(t-\tau) \\ &= \int_{-\infty}^{\infty} q(\tau)w(t-\tau)d\tau \end{aligned}$$

This is allowed as $q(\tau)$ is just a constant with respect to t.

We see now that we can obtain any response of the system by mixing the input function $q(t)$ with the unit impulse response function $w(t)$.

Definition 5.3.1: Convolution

The **convolution** of two functions $f(t)$ and $g(t)$ is defined as the integral of the product of the two functions, where one function is reversed and shifted. Mathematically, it is given by:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

Since integrals are linear operators, we also have:

$$\frac{d}{dt}(uv) = (p(D)u)v = u(p(D)v)$$

We also recall that we found it possible to "extract" the value of a function at a point by multiplying the function by a delta function and integrating, i.e.

$$\int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a)$$

We now saw that this was the convolution of $f(t)$ with $\delta(t)$.

Now, we can write the response of the system to the input $q(t)$ as the convolution of $q(t)$ with the unit impulse response function $w(t)$:

$$\begin{aligned} x(t) &= \int_{-\infty}^{\infty} q(\tau)w(t-\tau)d\tau \\ &= (q * w)(t) \end{aligned}$$

Example 5.3.1 (Using the convolution to solve a system)
Consider the system described by the following equation:

$$\dot{x} + 5x = \sin(\omega t)u(t)$$

Find the general solution to this system.

Solution: We recall that:

$$w(t) = u(t)e^{-kt}$$

In this case, we have that $k = 5$, so we can write the unit impulse response function as follows:

$$w(t) = u(t)e^{-5t}$$

So find $x(t)$, we simply convolve the input function $\sin(\omega t)u(t)$ with the unit impulse response function $w(t)$. We use complex replacement to find the convolution:

$$\begin{aligned} x(t) &= \int_{-\infty}^{\infty} \sin(\omega\tau)u(\tau)w(t-\tau)d\tau \\ &= \int_{0}^{t} \sin(\omega\tau)e^{-5(t-\tau)}d\tau \\ &= e^{-5t}\int_{0}^{t} \sin(\omega\tau)e^{5\tau}d\tau \\ &= e^{-5t} \cdot \frac{e^{5t}(5\sin(\omega t) - \omega\cos(\omega t)) + \omega}{25+\omega^2} \\ &= \frac{5\sin(\omega t) - \omega\cos(\omega t)}{25+\omega^2} + \frac{\omega e^{-5t}}{25+\omega^2} \end{aligned}$$

where $\omega \neq 5$. Notably, we have extracted both the particular solution (which is the second term) and the homogeneous solution (which is the first term) from the convolution. However, this formula does not work when resonance occurs, i.e. when $\omega = 5$.

5.3.1 Properties of convolution:

- Convolution is commutative: $f * g = g * f$
- Convolution is associative: $f * (g * h) = (f * g) * h$
- Convolution is distributive over addition: $f * (g + h) = f * g + f * h$

- Convolution with a delta function: $f * \delta = f$
- Convolution with a shifted delta function: $f * \delta(t-a) = f(t-a)$
- Convolution with a unit step function: $f * u = \int_{-\infty}^{t} f(\tau)d\tau$
- Convolution with an exponential function: $f * e^{at} = \int_{-\infty}^{t} f(\tau)e^{a(t-\tau)}d\tau$

5.4 Laplace transforms

The Laplace transform is a generalization of the above. Looking at the convolution product to get $x(t)$:

$$x(t) = \int_{-\infty}^{\infty} q(\tau)w(t-\tau)d\tau$$

We expand out $w(t-\tau)$ (for a first-order case, and assuming $t > 0$):

$$x(t) = \int_{-\infty}^{\infty} q(\tau)u(t-\tau)e^{-k(t-\tau)}d\tau$$

We can take out the e^{-kt} term since it is a constant with respect to τ:

$$x(t) = e^{-kt}\int_{-\infty}^{\infty} q(\tau)u(t-\tau)e^{k\tau}d\tau$$

Note that $u(t-\tau)$ is zero for $\tau > t$ and $q(\tau) = 0$ for $\tau < 0$, so we can change the limits of integration to get:

$$x(t) = e^{-kt}\int_{0}^{t} q(\tau)e^{k\tau}d\tau$$

Now we make the substitution $s = -k$ to get:

$$x(t) = e^{st}\int_{0}^{t} q(\tau)e^{-s\tau}d\tau$$

Now we can define the Laplace transform of $q(t)$ as follows:

$$\boxed{\mathcal{L}\{q(t)\} = \int_{0}^{\infty} q(\tau)e^{-s\tau}d\tau}$$

Then we can write $x(t)$ in terms of the Laplace transform of $q(t)$ as follows:

$$\begin{aligned} x(t) &= e^{st}\int_{0}^{t} q(\tau)e^{-s\tau}d\tau \\ &= e^{st}\mathcal{L}\{q(t)\} \end{aligned}$$

Definition 5.4.1: Laplace transform

The **Laplace transform** of a function $f(t)$ is defined as the integral of the product of the function and an exponential decay factor, given by:

$$\mathcal{L}\{f(t)\} = \int_{0}^{\infty} f(t)e^{-st}dt$$

where s is a complex number.

Recall that we found that $x(t) = (q * w)(t)$, where $w(t)$ is the unit impulse response function. When we take the Laplace transform of both sides, we get:

$$\begin{aligned} \mathcal{L}\{x(t)\} &= \mathcal{L}\{(q * w)(t)\} \\ &= \mathcal{L}\{q(t)\} \cdot \mathcal{L}\{w(t)\} \end{aligned}$$

Example 5.4.1 (Laplace transform of 1)
Find the Laplace transform of the function $f(t) = 1$.

Solution: We can compute the Laplace transform of $f(t) = 1$ as follows:

$$\begin{aligned}
\mathcal{L}\{1\} &= \int_0^\infty 1 \cdot e^{-st} dt \\
&= \int_0^\infty e^{-st} dt \\
&= \left[\frac{e^{-st}}{-s}\right]_0^\infty \\
&= \frac{1}{s}
\end{aligned}$$

Thus, we have that:

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s}}$$

Note:-

We know that $s \in \mathbb{C}$. Taking the modulus of e^{st}, we have:

$$\begin{aligned}
|e^{st}| &= |e^{(a+bi)t}| \\
&= |e^{at} \cdot e^{ibt}| \\
&= |e^{at}| \cdot |e^{ibt}| \\
&= e^{at} \cdot 1 \\
&= e^{at}
\end{aligned}$$

So the Laplace transform of $f(t) = 1$ converges if and only if $a < 0$, i.e. $\Re(s) < 0$.

Example 5.4.2 (Laplace transform of t^n)
Find the Laplace transform of the function $f(t) = t^n$, where n is a positive integer.

Solution: We can compute the Laplace transform of $f(t) = t^n$ as follows:

$$\begin{aligned}
\mathcal{L}\{t^n\} &= \int_0^\infty t^n e^{-st} dt \\
&= \left[-\frac{t^n e^{-st}}{s}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1} e^{-st} dt \\
&= 0 + \frac{n}{s}\mathcal{L}\{t^{n-1}\}
\end{aligned}$$

Now we can apply this process iteratively to find the Laplace transform of t^n:

$$\begin{aligned}
\mathcal{L}\{t^n\} &= \frac{n}{s}\mathcal{L}\{t^{n-1}\} \\
\mathcal{L}\{t^n\} &= \frac{n}{s}\mathcal{L}\{t^{n-1}\} \\
&= \frac{n}{s}\cdot\frac{n-1}{s}\mathcal{L}\{t^{n-2}\} \\
&= \frac{n}{s}\cdot\frac{n-1}{s}\cdot\frac{n-2}{s}\mathcal{L}\{t^{n-3}\} \\
&\vdots \\
&= \frac{n(n-1)(n-2)\cdots 1}{s^n}\mathcal{L}\{t^0\} \\
&= \frac{n!}{s^n}\mathcal{L}\{1\}
\end{aligned}$$

It remains to find $\mathcal{L}\{1\}$:

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\,dt = \left[-\frac{e^{-st}}{s}\right]_0^\infty = \frac{1}{s}$$

Therefore:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^n}\cdot\frac{1}{s} = \frac{n!}{s^{n+1}}$$

Laplace transforms also satisfy linearity, i.e. for any constants a and b, we have:

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

They also satisfy shifting properties, i.e. for any constant a, we have:

$$\begin{aligned}
\mathcal{L}\{e^{at}f(t)\} &= F(s-a) \\
\mathcal{L}\{f(t-a)u(t-a)\} &= e^{-as}F(s)
\end{aligned}$$

5.5 Using Laplace transforms to solve linear systems

Taking the Laplace transform of a derivative:

$$\begin{aligned}
\mathcal{L}\{\dot{x}(t)\} &= \int_0^\infty \dot{x}(t)e^{-st}dt \\
&= \left[x(t)e^{-st}\right]_0^\infty + s\int_0^\infty x(t)e^{-st}dt \\
&= 0 - x(0) + sX(s) \\
&= sX(s) - x(0)
\end{aligned}$$

Taking the Laplace transform of a second derivative:

$$\begin{aligned}
\mathcal{L}\{\ddot{x}(t)\} &= \int_0^\infty \ddot{x}(t)e^{-st}dt \\
&= \left[\dot{x}(t)e^{-st}\right]_0^\infty + s\int_0^\infty \dot{x}(t)e^{-st}dt \\
&= 0 - \dot{x}(0) + s(sX(s) - x(0)) \\
&= s^2X(s) - sx(0) - \dot{x}(0)
\end{aligned}$$

So we see that the Laplace transform of a derivative is a polynomial in s multiplied by the Laplace transform of the original function, minus some terms involving the initial conditions.

Function	$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
Unit impulse	$\delta(t)$	1
Unit step	$u(t)$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Power	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{at}	$\frac{1}{s-a}$
Sine	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
Cosine	$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
Damped sine	$e^{at}\sin(\omega t)$	$\frac{\omega}{(s-a)^2+\omega^2}$
Damped cosine	$e^{at}\cos(\omega t)$	$\frac{s-a}{(s-a)^2+\omega^2}$
Hyperbolic sine	$\sinh(at)$	$\frac{a}{s^2-a^2}$
Hyperbolic cosine	$\cosh(at)$	$\frac{s}{s^2-a^2}$

Table 5.1: Common Laplace Transform Pairs

Example 5.5.1 (Using Laplace transforms to solve a system)
Consider the system described by the following equation:

$$\dot{w} + 3w = \delta(t)$$

with initial rest conditions (i.e. $w(0) = 0$ and $\dot{w}(0) = 0$).
Find the response of the system to this input.

Solution: Taking the Laplace transform of both sides of the equation, we get:

$$\begin{aligned} sW(s) - w(0) + 3W(s) &= 1 \\ (s+3)W(s) &= 1 \\ W(s) &= \frac{1}{s+3} \end{aligned}$$

Note that $p(s) = s + 3$, so $W(s) = \frac{1}{p(s)}$.

We need to find some function $w(t)$ such that $\mathcal{L}\{w(t)\} = \frac{1}{s+3}$. We can use the shifting property of Laplace transforms to find $w(t)$:

$$\begin{aligned} \mathcal{L}\{e^{-3t}\} &= \frac{1}{s-(-3)} \\ &= \frac{1}{s+3} \end{aligned}$$

Thus, we have that:

$$\boxed{w(t) = e^{-3t}u(t)}$$

Interestingly, this demonstrates that for the unit impulse response function, we can extract $p(s)$ from $W(s)$.

Property	Transform
Linearity	$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$
First derivative	$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$
Second derivative	$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$
n-th derivative	$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$
Integration	$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{F(s)}{s}$
Frequency shift	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Time shift	$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$
Scaling	$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$
Convolution	$\mathcal{L}\{f(t) * g(t)\} = F(s) \cdot G(s)$
Multiplication by t	$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^nF(s)}{ds^n}$

Table 5.2: Laplace Transform Properties

Example 5.5.2

Consider the system described by the following equation:

$$\dot{x} + 3x = e^{-t}, \quad x(0^+) = 0$$

Solution: Taking the Laplace transform of both sides of the equation, we get:

$$\begin{aligned} sX(s) - x(0) + 3X(s) &= \frac{1}{s+1} \\ (s+3)X(s) &= \frac{1}{s+1} \\ X(s) &= \frac{1}{(s+3)(s+1)} \end{aligned}$$

To find $x(t)$, we can use partial fraction decomposition to write $X(s)$ as a sum of simpler fractions:

$$\begin{aligned} X(s) &= \frac{1}{(s+3)(s+1)} \\ &= \frac{A}{s+3} + \frac{B}{s+1} \end{aligned}$$

Solving for A and B, we get:

$$\begin{aligned} A(s+1) + B(s+3) &= 1 \\ A = \frac{1}{2}, \quad B &= -\frac{1}{2} \end{aligned}$$

Then, we can write $X(s)$ as follows:

$$X(s) = \frac{1/2}{s+3} - \frac{1/2}{s+1}$$

We can use e^{at} to find the inverse Laplace transform of each term:

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$$
$$\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$$

Thus, we have that:

$$\boxed{x(t) = \frac{1}{2}e^{-3t}u(t) - \frac{1}{2}e^{-t}u(t)}$$

We can also get both $x_h(t)$ and $x_p(t)$ from this. In particular, the homogeneous solution is the term with e^{-3t}, and the particular solution is the term with e^{-t}.

Example 5.5.3

Consider the system described by the following equation:

$$\dot{x} + 3x = e^{-t}, \quad x(0^+) = 5$$

Solution: We interpret this as the system being at rest before $t = 0$, and then at $t = 0$, the system is given an initial condition of $x(0^+) = 5$.

So we rewrite as:

$$\dot{x} + 3x = e^{-t} + 5\delta(t)$$

Then we can take the Laplace transform of both sides to get:

$$\begin{aligned} sX(s) - x(0) + 3X(s) &= \frac{1}{s+1} + 5 \\ (s+3)X(s) &= \frac{1}{s+1} + 5 \\ X(s) &= \frac{1}{(s+3)(s+1)} + \frac{5}{s+3} \end{aligned}$$

Then we can use partial fraction decomposition to find the inverse Laplace transform of the first term, and we can use e^{at} to find the inverse Laplace transform of the second term:

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$$
$$\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$$

Thus, we have that:

$$\boxed{x(t) = \frac{1}{2}e^{-3t}u(t) - \frac{1}{2}e^{-t}u(t) + 5e^{-3t}u(t)}$$

Example 5.5.4 (n-th order system)

Consider a function $f(t)$ such that $\mathcal{L}\{f(t)\} = F(s)$. Find the Laplace transform of $f^{(n)}(t)$, where $f^{(n)}(t)$ is the n-th derivative of $f(t)$.

Solution: We can compute the Laplace transform of $f^{(n)}(t)$ as follows:

$$\begin{aligned} \mathcal{L}\{f^{(n)}(t)\} &= \int_0^\infty f^{(n)}(t)e^{-st}dt \\ &= \left[f^{(n-1)}(t)e^{-st}\right]_0^\infty + s\int_0^\infty f^{(n-1)}(t)e^{-st}dt \\ &= 0 - f^{(n-1)}(0) + s\mathcal{L}\{f^{(n-1)}(t)\} \\ &= s\mathcal{L}\{f^{(n-1)}(t)\} - f^{(n-1)}(0) \end{aligned}$$

Continuing this process iteratively, we get:

$$\begin{aligned}
\mathcal{L}\{f^{(n)}(t)\} &= s\mathcal{L}\{f^{(n-1)}(t)\} - f^{(n-1)}(0) \\
&= s(s\mathcal{L}\{f^{(n-2)}(t)\} - f^{(n-2)}(0)) - f^{(n-1)}(0) \\
&= s^2\mathcal{L}\{f^{(n-2)}(t)\} - sf^{(n-2)}(0) - f^{(n-1)}(0) \\
&\vdots \\
&= s^n\mathcal{L}\{f(t)\} - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0) \\
&= s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)
\end{aligned}$$

Read this as $s^n F(s)$ minus a sum of terms involving the initial conditions of $f(t)$ and its derivatives up to order $n-1$.

Example 5.5.5 (Solving a second-order system)
Consider the following equation:

$$\ddot{w} + 2\dot{w} + 2w = \delta(t)$$

Find the impulse response function of this system.

Solution: Taking the Laplace transform of both sides of the equation, we get:

$$\begin{aligned}
s^2W(s) - sw(0) - \dot{w}(0) + 2(sW(s) - w(0)) + 2W(s) &= 1 \\
(s^2 + 2s + 2)W(s) &= 1 \\
W(s) &= \frac{1}{s^2 + 2s + 2}
\end{aligned}$$

To find $w(t)$, we can complete the square in the denominator of $W(s)$:

$$W(s) = \frac{1}{(s+1)^2 + 1}$$

First, from the Laplace table, we see a s^2 term indicating a sinusoidal function. Then we can use the shifting property of Laplace transforms to find $w(t)$:

$$\mathcal{L}\{e^{-t}\sin t\} = \frac{1}{(s+1)^2 + 1}$$

Thus, we have that:

$$\boxed{w(t) = e^{-t}\sin t \cdot u(t)}$$

Observe the $u(t)$ term, which comes from the fact that the system is at rest before $t = 0$.

Definition 5.5.1: Weight function

The **weight function** of a system is the response of the system to a unit impulse input ($\delta(t)$).

Definition 5.5.2: Transfer function

The **transfer function** of a system is the Laplace transform of the weight function of the system.

In particular, the transfer function of $P(D)$ for the system:

$$P(D)x = q(t)$$

is given by passing a unit impulse input through the system and taking the Laplace transform of the output. So we have that:

$$\begin{aligned} p(D)w(t) &= \delta(t) \\ \mathcal{L}\{p(D)w(t)\} &= \mathcal{L}\{\delta(t)\} \\ p(s)W(s) &= 1 \\ W(s) &= \frac{1}{p(s)} \end{aligned}$$

With this, we can see that to find $x(t)$, we can first find the transfer function $W(s)$, then find the Laplace transform of the input $q(t)$, and then multiply these two together to get $X(s)$, and finally take the inverse Laplace transform to get $x(t)$.

Example 5.5.6 (Using the transfer function to solve a system)
Consider the system described by the following equation:

$$\ddot{x} + 4x = \cos(2t)u(t)$$

Note that $u(t)$ signals rest initial conditions. Find the response of the system to this input.

Solution: First, we find the transfer function of the system. We can write the system as follows:

$$\begin{aligned} \ddot{w} + 4w &= \delta(t) \\ p(D)w &= \delta(t) \end{aligned}$$

Recall that $W(s) = \frac{1}{p(s)}$, so we need to find $p(s)$:

$$p(s) = s^2 + 4$$

Thus, we have that:

$$W(s) = \frac{1}{s^2 + 4}$$

Now, multiplying $W(s)$ by the Laplace transform of the input $\cos(2t)u(t)$, we get:

$$\begin{aligned} X(s) &= W(s) \cdot \mathcal{L}\{\cos(2t)u(t)\} \\ &= \frac{1}{s^2 + 4} \cdot \frac{s}{s^2 + 4} \\ &= \frac{s}{(s^2 + 4)^2} \end{aligned}$$

Note this is equivalent to taking the convolution of the input with the weight function of the system, and then taking the Laplace transform of the result.

We now need to find the inverse Laplace transform. Using the table, we can see that the inverse Laplace transform of $\frac{1}{s^2+4}$ is $\sin(2t)$.

However, $X(s)$ looks something like the derivative of $\frac{1}{s^2+4}$. In particular:

$$\begin{aligned} \frac{d}{ds}\left(\frac{1}{s^2 + 4}\right) &= -\frac{2s}{(s^2 + 4)^2} \\ -\frac{1}{2} \cdot \frac{d}{ds}\left(\frac{1}{s^2 + 4}\right) &= \frac{s}{(s^2 + 4)^2} \end{aligned}$$

So we can use the property of Laplace transforms that states that the Laplace transform of $tf(t)$ is equal to $-F'(s)$, where $F(s)$ is the Laplace transform of $f(t)$. In particular, we have that:

$$\mathcal{L}\{t\sin(2t)\} = \frac{4s}{(s^2 + 4)^2}$$

scaling by $\frac{1}{4}$ gives

$$\frac{1}{4} \cdot \frac{4s}{(s^2 + 4)^2} = \frac{s}{(s^2 + 4)^2}$$

Thus, we have that:

$$\boxed{x(t) = \frac{1}{4} t \sin(2t) \cdot u(t)}$$

5.6 Poles

We see that $F(s) = \frac{1}{s}$ has a singularity where $s = 0$, at which the function "blows up". We call such a point a **pole** of the function. In general, a pole of a function is a point in the complex plane where the function becomes unbounded (i.e., it goes to infinity).

Example 5.6.1 (Poles of a function)
Find the poles of the function $F(s) = \frac{1}{s^2+4}$.

Solution: The poles of $F(s)$ are the values of s for which the denominator is zero:

$$s^2 + 4 = 0 \implies s^2 = -4 \implies s = \pm 2i$$

Thus, the poles are $s = 2i$ and $s = -2i$.

In the complex plane, the poles of a function have two parts:

- The real part of the pole, which determines the exponential growth or decay of the corresponding term in the time domain.
- The imaginary part of the pole, which determines the oscillatory behavior of the corresponding term in the time domain.

Poles with $\text{Re}(s) < 0$ correspond to terms that decay exponentially, while poles with $\text{Re}(s) > 0$ correspond to terms that grow exponentially. Poles with $\text{Re}(s) = 0$ correspond to terms that neither grow nor decay, but may oscillate if they have a nonzero imaginary part.

Definition 5.6.1: Stability

A system is **stable** if all poles of its transfer function have negative real parts. This means that the system's response to any bounded input will eventually decay to zero as time goes to infinity.

Note:-

A note on Fourier transforms as related to Laplace transforms. A continuous Fourier transform:

$$\mathbb{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

is a special case of the Laplace transform where $s = i\omega$. So Laplace transforms can be thought of as a generalization of Fourier transforms, where we allow for complex values of s that can have both real and imaginary parts. This allows us to analyze systems that may not be purely oscillatory, but may also have exponential growth or decay.

Chapter 6

Systems of ODEs

6.1 First-order systems of ODEs

Consider the following:

$$\ddot{x} + x\dot{x} + \sin x = 0$$

This is a 2nd order, non-linear ODE. We can think of this as a system of ODEs by letting $y = \dot{x}$, so $\dot{y} = \ddot{x}$. Then we can rewrite the above as:

$$\begin{cases} \dot{x} = y \\ \dot{y} = -xy - \sin x \end{cases}$$

While this is a first-order system of ODEs, the solution is now two-dimensional, so we can represent it as a vector:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} y \\ -xy - \sin x \end{bmatrix}$$

We call this a **phase space** representation of the system. The solution to this system will be a curve in the xy-plane, which we can visualize as a trajectory of the system.

Definition 6.1.1: Linear system of ODEs

A system of ODEs is linear if it can be written in the form:

$$\dot{u} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Letting the matrix be A, we can write a linear system of ODEs as:

$$\boxed{\dot{u} = Au}$$

Note:-

If A was a 1x1 matrix, then this would be a scalar and hence a single ode of the form:

$$\dot{u} = \lambda u$$

which we solve as:

$$u(t) = u(0)e^{\lambda t}$$

Now, let $A \in \mathbb{R}^{m \times n}$, and let $u(t) \in \mathbb{R}^n$ be a vector of functions. Then we can write the system of ODEs as:

$$\dot{u}(t) = Au(t)$$

We guess that the solution is of the form:

$$u(t) = e^{At}u(0)$$

However, we must define what e^{At} means.
Recall the Taylor series expansion of the exponential function:

$$e^{\lambda t} = \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!}$$

Then we can define the matrix exponential as:

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}$$

We take the first derivative of e^{At}:

$$\frac{d}{dt}e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^{k-1}}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = Ae^{At}$$

Thus, we have:

$$u(t) = e^{At}u(0)$$

is indeed a solution to the system of ODEs.

In particular, we found that $\dot{u}(t) = Ae^{At}u(0) = Au(t)$, which is exactly the system of ODEs we wanted to solve.

Example 6.1.1 (Trivial example)
Let $\dot{u} = Au$ where:

$$A = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$$

Then we let:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ dy \end{bmatrix}$$

Then solving each equation separately, we get:

$$x(t) = x(0)e^{at}, \quad y(t) = y(0)e^{dt}$$

$$\Rightarrow u(t) = \begin{bmatrix} x(0)e^{at} \\ y(0)e^{dt} \end{bmatrix} = e^{At}u(0)$$

where

$$u(0) = \begin{bmatrix} x(0) \\ y(0) \end{bmatrix}$$

Consider each of the cases:

- $a = d < 0$: then the solution decays to zero as $t \to \infty$. Stability is achieved in both cases.
- $a = d > 0$: then the solution grows exponentially as $t \to \infty$. Stability is never achieved.
- $a < d < 0$: then the solution decays to zero as $t \to \infty$, but the y component decays faster than the x component. Stability is achieved in both cases.
- $d < a < 0$: then the solution decays to zero as $t \to \infty$, but the x component decays faster than the y component. Stability is achieved in both cases.
- $a < 0 < d$: then the solution decays to zero as $t \to \infty$, but the x component decays while the y component grows. Stability is never achieved.
- $d < 0 < a$: then the solution decays to zero as $t \to \infty$, but the y component decays while the x component grows. Stability is never achieved.

6.2 General solution to linear systems of ODEs

Let $\dot{u} = Au$ where $u = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $A \in \mathbb{R}^{n \times n}$.

We will use the eigenvalue decomposition of A to find the general solution to this system of ODEs. Let $A = PDP^{-1}$ where D is a diagonal matrix containing the eigenvalues of A, and P is a matrix whose columns are the corresponding eigenvectors.

Recall the eigenvector v corresponding to an eigenvalue λ satisfies:

$$Av = \lambda v$$

Then we have:

$$(A - \lambda I)v = 0$$

We find that this is equivalent to stating that $\det(A - \lambda I) = 0$ (remember that the determinant is zero if and only if the matrix is singular, which is the case when there is a non-trivial solution to the above equation).

Then, solving:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} \\ &= 0 \\ &= \prod_{i=1}^{n} (\lambda_i - \lambda) = 0 \end{aligned}$$

For a two-dimensional system, we see that the trace and determinant appear in the characteristic polynomial:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) \\ &= \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0 \end{aligned}$$

This is the characteristic polynomial of A. Furthermore, we see that the behaviour of the system is characterized by the difference between the trace and determinant of A.

The exact relationship is given by:

$$\lambda_{1,2} = \frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2 - 4\det(A)}}{2}$$

We have:

- If $\operatorname{tr}(A)^2 - 4\det(A) > 0$, then the eigenvalues are real and distinct, and the system is a node.
- If $\operatorname{tr}(A)^2 - 4\det(A) = 0$, then the eigenvalues are real and repeated, and the system is a degenerate node.
- If $\operatorname{tr}(A)^2 - 4\det(A) < 0$, then the eigenvalues are complex conjugates, and the system is a spiral.

And, we saw that the solution to the system of ODEs when D is diagonal is given by:

$$u(t) = e^{At}u(0) = Pe^{Dt}P^{-1}u(0)$$

6.2.1 Steps to solve a system of ODEs

1. Find the eigenvalues and eigenvectors of A.
2. Form the matrix P whose columns are the eigenvectors and the diagonal matrix D whose diagonal entries are the corresponding eigenvalues.
3. Compute P^{-1}.
4. The general solution is given by $u(t) = Pe^{Dt}P^{-1}u(0)$.

Example 6.2.1 (Example of solving a system of ODEs)
Let a system of ODEs be given by:

$$\begin{cases} \dot{x} = x + y \\ \dot{y} = 4x - 2y \end{cases}$$

with initial conditions $x(0) = 2$ and $y(0) = -3$.
We can write this system in matrix form as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Then, we find the eigenvalues of A by using the trace and determinant:

$$\operatorname{tr}(A) = 1 + (-2) = -1, \quad \det(A) = (1)(-2) - (4)(1) = -6$$

Computing with the characteristic polynomial, we have:

$$\begin{aligned} \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) &= 0 \\ \lambda^2 + \lambda - 6 &= 0 \\ (\lambda + 3)(\lambda - 2) &= 0 \end{aligned}$$

Then the eigenvalues are $\lambda_1 = -3$ and $\lambda_2 = 2$.
For $\lambda_1 = -3$, we have:

$$\begin{aligned} \begin{bmatrix} 1 - (-3) & 1 \\ 4 & -2 - (-3) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} &= 0 \\ \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} &= 0 \end{aligned}$$

If we row-reduce this matrix, we get:

$$\begin{bmatrix} 1 & \frac{1}{4} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

This gives us the eigenvector corresponding to λ_1 as $v_1 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$.

For $\lambda_2 = 2$, we have:

$$\begin{aligned} \begin{bmatrix} 1 - 2 & 1 \\ 4 & -2 - 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} &= 0 \\ \begin{bmatrix} -1 & 1 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} &= 0 \end{aligned}$$

Row-reducing this matrix, we get:

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

This gives us the eigenvector corresponding to λ_2 as $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
Then, we can form the matrix P and the diagonal matrix D as follows:

$$P = \begin{bmatrix} -1 & 1 \\ 4 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$$

Rather than compute P^{-1}, we can write the solution as:

$$u(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} e^{-3t}$$

Then, we can use the initial conditions to solve for c_1 and c_2:

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} c_1 - c_2 \\ c_1 + 4c_2 \end{bmatrix}$$

This gives us the system of equations:

$$c_1 - c_2 = 2$$
$$c_1 + 4c_2 = -3$$

Solving this system, we get:

$$c_1 = 1$$
$$c_2 = -1$$

Thus, the solution to the system of ODEs is given by:

$$u(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} - \begin{bmatrix} -1 \\ 4 \end{bmatrix} e^{-3t}$$

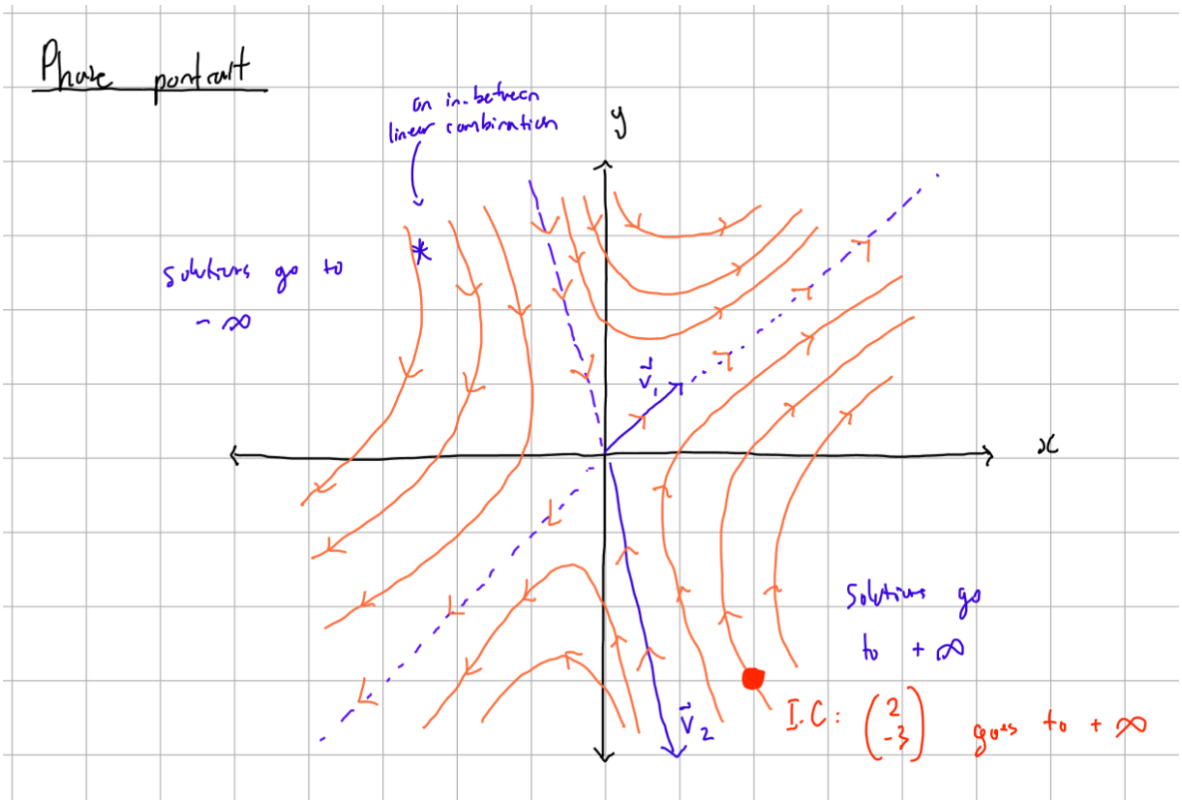
Note that computing $c_1 v_1 + c_2 v_2 = u(0)$ where v_1 and v_2 are eigenvectors is equivalent to inverting the matrix P then multiplying by $u(0)$, which is the same as computing $P^{-1}u(0)$.
Also, the eigenvalues tell us about the stability of the system. Since $\lambda_1 = -3 < 0$ and $\lambda_2 = 2 > 0$, we see that the system is unstable, as one of the eigenvalues is positive. The solution will grow exponentially as $t \to \infty$ due to the term involving e^{2t}, which corresponds to the eigenvalue λ_2.

6.3 Phase portraits

Consider the system of ODEs given in the previous example with $\lambda_1 = 2 > 0$ and $\lambda_2 = -3 < 0$. We found the eigenvectors:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

Then we can draw a phase portrait by plotting the eigenvectors and the trajectories of the system in the xy-plane. The eigenvector corresponding to the positive eigenvalue will indicate the direction of growth, while the eigenvector corresponding to the negative eigenvalue will indicate the direction of decay.



Note that the characteristic polynomial $\lambda^2 + \lambda - 6 = p(\lambda)$ is exactly the same as the one-dimensional 2nd order system:

$$\begin{aligned}
\ddot{x} = \frac{d}{dt}\dot{x} &= \frac{d}{dt}(x + y) \\
&= \dot{x} + \dot{y} \\
&= \dot{x} + 4x - 2y \\
&= \dot{x} + 4x - 2(\dot{x} - x) \\
&= -\dot{x} + 6x
\end{aligned}$$

Then we have that $\ddot{x} + \dot{x} - 6x = 0$ has the same characteristic polynomial as the system of ODEs, and hence the same eigenvalues. This is a general result: the characteristic polynomial of a system of ODEs is the same as the characteristic polynomial of the corresponding higher-order ODE.

Chapter 7

Harmonic oscillators

Consider the second-order ODE given by:

$$\ddot{x} + \omega^2 x = 0$$

By using a change of variables $y = \dot{x}$, we can rewrite this as a system of ODEs:

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\omega^2 x \end{cases}$$

This system of ODEs can be represented in matrix form as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega^2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The characteristic polynomial of the matrix is given by:

$$\det \begin{bmatrix} -\lambda & 1 \\ -\omega^2 & -\lambda \end{bmatrix} = \lambda^2 + \omega^2 = 0$$

This gives us the eigenvalues $\lambda = \pm i\omega$. We find that the eigenvectors are:

$$v_1 = \begin{bmatrix} 1 \\ i\omega \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -i\omega \end{bmatrix}$$

Then, we can write the general solution to the system of ODEs as:

$$u(t) = c_1 e^{i\omega t} v_1 + c_2 e^{-i\omega t} v_2$$

We have a real solution, so the coefficients c_1 and c_2 must be complex. Using initial conditions $u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, we solve:

$$\begin{aligned} \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= c_1 \begin{bmatrix} 1 \\ i\omega \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -i\omega \end{bmatrix} \\ &= \begin{bmatrix} -i(c_1 - c_2) \\ \omega(c_1 + c_2) \end{bmatrix} \end{aligned}$$

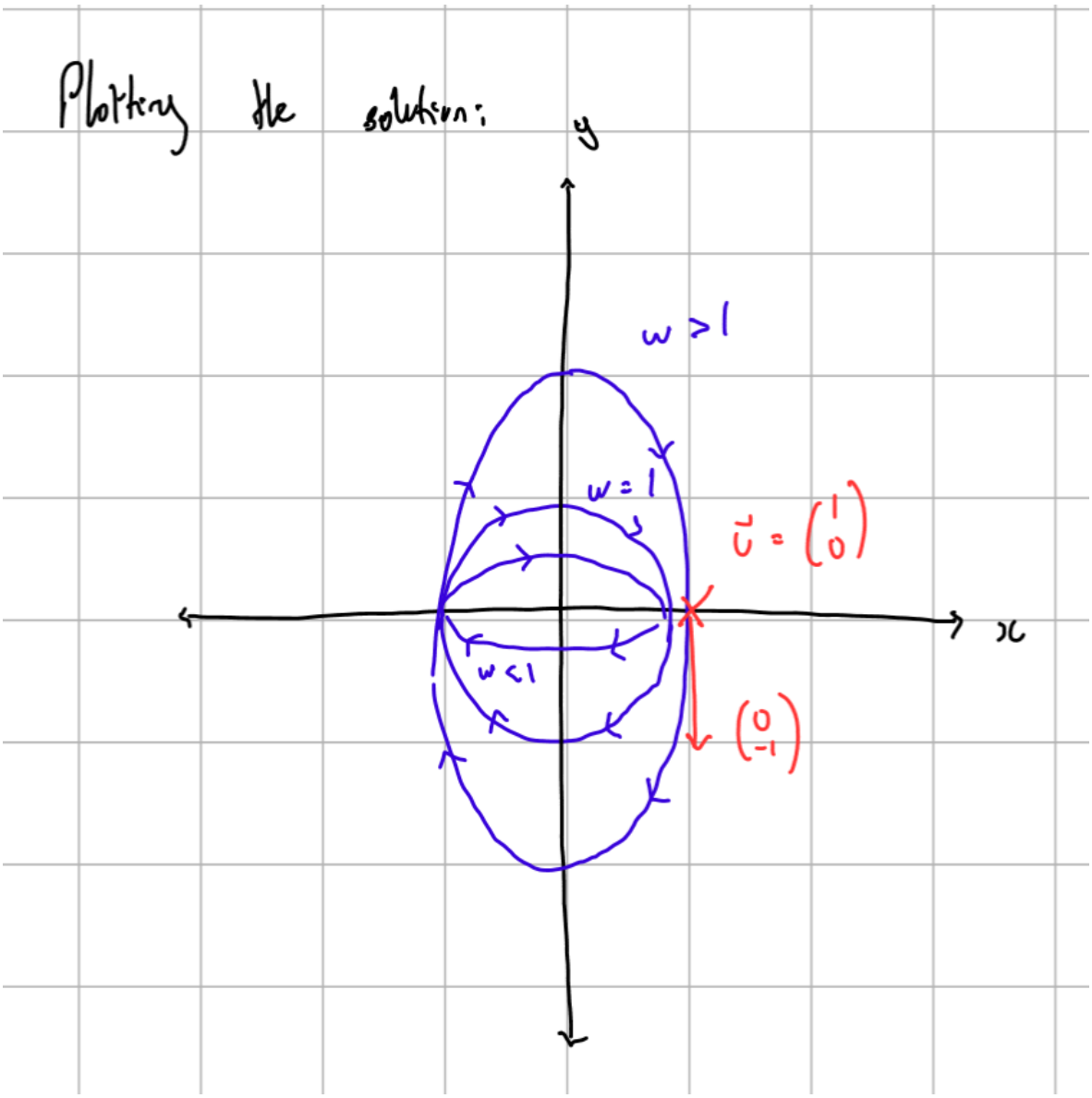
Then from $c_1 = -c_2$, we get $1 = -i(2c_2)$, so $c_2 = \frac{i}{2}$ and $c_1 = -\frac{i}{2}$. Thus, the solution to the system of ODEs is given by:

$$u(t) = \frac{i}{2} e^{-i\omega t} v_2 - \frac{i}{2} e^{i\omega t} v_1$$

Using Euler's formula, we can rewrite this as:

$$u(t) = \begin{bmatrix} \cos(\omega t) \\ -\omega \sin(\omega t) \end{bmatrix}$$

Now, plotting the phase portait of this system, we get cycles around the origin.



Example 7.0.1 (Love affair)

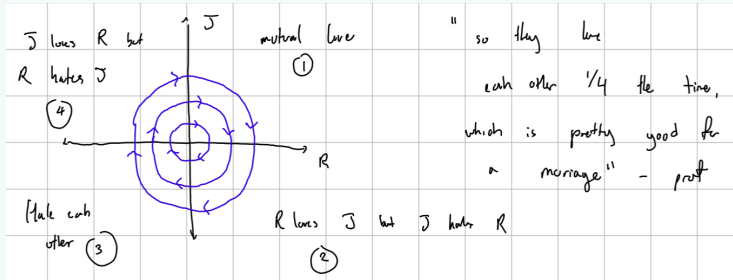
Consider two functions:

$$\begin{cases} R(t) = \text{Romeo's love for Juliet at time } t \\ J(t) = \text{Juliet's love for Romeo at time } t \end{cases}$$

We first consider when the love that Romeo has for Juliet depends only on Juliet's love for Romeo, and vice versa. Let $a, b > 0$, then:

$$\begin{cases} \dot{R} = aJ \\ \dot{J} = -bR \end{cases}$$

This system describes a situation where Romeo loves Juliet the more she loves him, but Juliet loves Romeo the less he loves her. We see that this is equivalent to the harmonic oscillator system above (with different coefficients), so we would observe cycling behaviour.



Now, we consider the general case:

$$\begin{cases} \dot{R} = aR + bJ \\ \dot{J} = cR + dJ \end{cases}$$

The choice of sign for each coefficient signals the romantic style.
Consider a case where $a < 0$, $b > 0$, and we have:

$$\begin{cases} \dot{R} = aR + bJ \\ \dot{J} = bR + aJ \end{cases}$$

Then the matrix of this system is given by:

$$A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

It should be easy to see that the eigenvalues of this matrix are given by $\lambda_1 = a + b$ and $\lambda_2 = a - b$. Also, the eigenvectors are:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

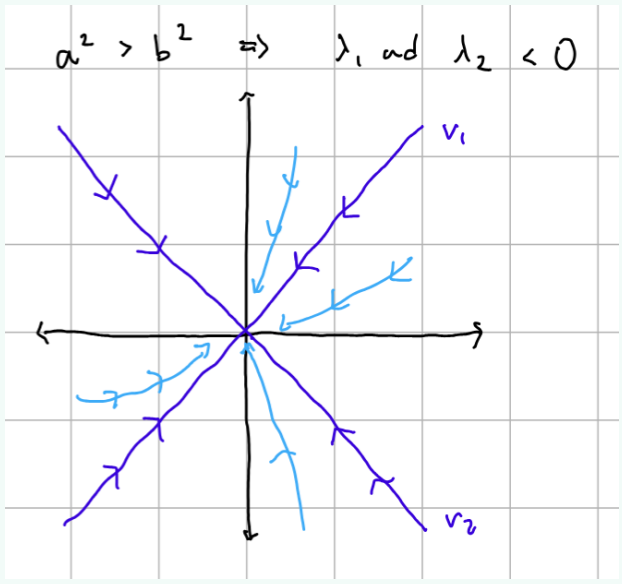
We can also compute the determinant:

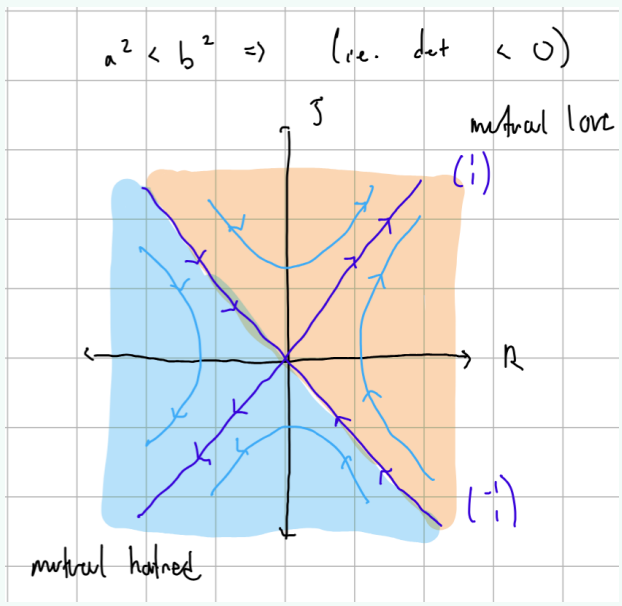
$$\det(A) = a^2 - b^2$$

Then depending on the value of a and b, we can have different behaviours:

- If $a^2 > b^2$, then the eigenvalues are real and negative, and the system is a stable node. The love affair will eventually die out.

- If $a^2 < b^2$, then the eigenvalues are real and of opposite signs, and the system is a saddle. The love affair will be unstable, with one person loving the other more and more while the other person loves them less and less.





7.0.1 Repeated eigenvalues

Given the system $\dot{u} = Au$ (for a 2-dimensional system), if A has a repeated eigenvalue, then:

- Then only one eigenvector v_1 exists.
- There is only one fundamental solution given by: $u_1(t) = e^{\lambda t} v_1$ where v_1 is the eigenvector.

  We need another solution which is linearly independent from v_1.

7.0.2 Finding a second solution when there is a repeated eigenvalue

Let v_1 be the eigenvector corresponding to the repeated eigenvalue λ. We can find a second solution of the form:

$$u_2(t) = e^{\lambda t}(tv_1 + w)$$

where w is a vector that we need to determine.

Plug $u_2(t)$ into $\dot{u} = Au$:

$$\begin{aligned} \dot{u}_2(t) &= Au_2(t) \\ e^{\lambda t}(v_1 + t\lambda v_1 + \lambda w) &= e^{\lambda t}(tAv_1 + Aw) \\ v_1 + t\lambda v_1 + \lambda w &= t\lambda v_1 + Aw \quad \text{using } Av_1 = \lambda v_1 \\ v_1 + \lambda w &= Aw \end{aligned}$$

Rearranging this, we get:

$$(A - \lambda I)w = v_1$$

Since v_1 is an eigenvector corresponding to λ, we know that $(A - \lambda I)v_1 = 0$. Thus, the matrix $A - \lambda I$ is singular, and we can find a solution for w using methods such as Gaussian elimination or by finding the null space of $A - \lambda I$.

The general solution to the system of ODEs in this case is given by:

$$\boxed{u(t) = c_1 e^{\lambda t} v_1 + c_2 e^{\lambda t}(tv_1 + w)}$$

Example 7.0.2 (Example of repeated eigenvalues)

Consider the system of ODEs given by:

$$\begin{cases} \dot{x} = 2x + y \\ \dot{y} = -4x - 2y \end{cases}$$

We can write this system in matrix form as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

We will find that $\lambda = 0$ is a repeated eigenvalue, and that $u_1(t) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is the corresponding eigenvector.

Then, we can find w by solving:

$$\begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

We can row-reduce this matrix to find a solution for w:

$$\begin{bmatrix} 2 & 1 & 1 \\ -4 & -2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

This gives us: $2a + b = 1$. We could choose $a = 0$ and $b = 1$, then:

$$w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Thus, the general solution to the system of ODEs is given by:

$$u(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \left(t \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$